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Installation Manual for

Vessel Remote Server, VRS 002 G2

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1 Scope and purpose

1.1 References

DBS12436 VRS 002 File Transport System REST/JSON API
MAN13426 User manual for the VRS password and security tool
MAN13735 VRS remote management TeamViewer manual

1.2 Terms and Abbreviations

DFS Dump for Support: A report which may be used for troubleshooting by Danelec Marine.

OPT Operational Performance Test. A test carried out for a VDR installation using the bridge control panel. The purpose of the test is to display the decoded and calibrated values for the recorded data.

VRS Vessel Remote Server

URL Uniform Resource Locator

1.2.1 Special terms used for the description of the VRS

Data processor: An application in the VRS which can generate reports.

Data processor configuration: A configuration for a data processor.

Data processor configuration file: A file containing a configuration for a data processor. A data processor configuration is created using the VDR Explorer.

Report: A file containing a data set made by a data processor.

Report configuration: A data processor configuration may contain one or more report configurations. A report configuration defines the data set, transmission interval and destination for a report.

Report Name: A unique ID for a report within a data processor configuration.

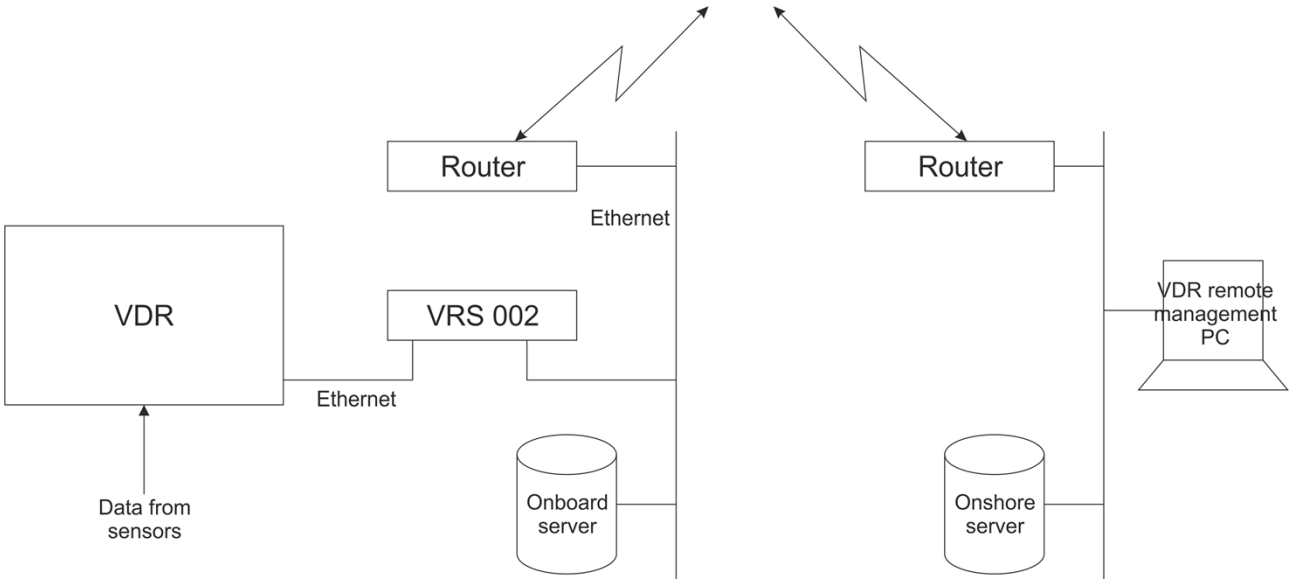
Destination: Destination is the generic term for where to send a report. The main parameter for a destination is typically the IP address of a server.

Destination name: Destination name is the common reference for a destination used in the VRS configuration, VDR Explorer GUI, and in a report configuration.

2 System overview

The VRS 002 G2 may currently be used for two independent applications.

- Remote VDR management of Danelec Marine VDRs.
- Transmission of telemetry data (reports) to systems onshore and onboard the vessel.



System overview

2.1 Remote VDR management

The VRS 002 G2 contains a gateway for the VDR remote management tool. The VDR remote management tool is an application for a PC, typically installed onshore.

VDR remote management includes:

- Alert monitoring
- Pre-APT
- Easy download of data to shore following an incident
- Remote configuration of the VDR
- Easy generation of reports which may be used for troubleshooting (DFS)

2.2 Transmission of telemetry data

The VRS 002 G2 is able the generate reports (files) which contain telemetry data and send them to servers and other systems. The data is typically derived from the data recorded by the VDR, but the VRS 002 G2 may also use other data sources.

3 Installation

The VRS 002 G2 must be mounted on a surface away from direct sunlight and water spray, IEC 60945 protected. For installation drawings, see section 18.

3.1 Installation, motion-tracking sensor

If the built-in motion-tracking sensor is going to be used, the VRS 002 G2 must be installed horizontally and parallel with the vessel's centerline. Minor deviations may be offset using the configuration interface in the VRS.

All of the 24 possible orientations are supported. For example, if the lid is facing up, the front may be facing starboard, port, stern, or the bow.

3.2 Hardware specification

Interfaces:
Five IEEE 802.1 1000BaseT/100BaseTX (Ethernet) interface
One RS-422 serial interface
One digital input (3-24VDC)

USB 3.0 port (currently not utilized)

Power:
12-24 VDC IEC 60945 nominal, 10W.
110V-230VAC IEC 60945 nominal, 12W using the AC adaptor supplied with the product.

Approval:
IEC 60945 protected.

SWAP:
M2 SSD (cuurenly not supported)

Options
1TB M2 SSD for additional storage of data.

Dimensions:
See section 18, Installation drawings.

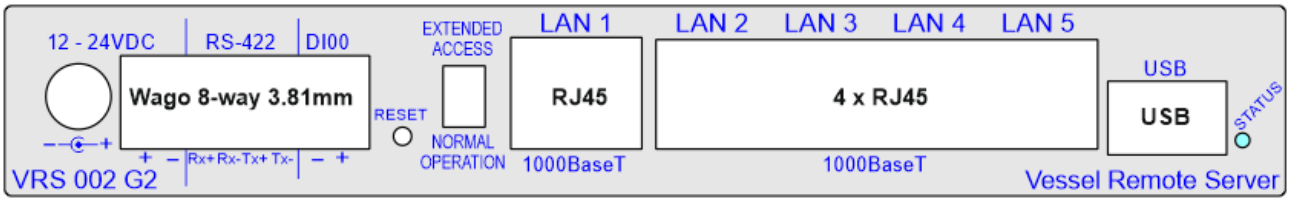
3.3 Power requirements

Power consumption – 10W(VDC) / 12W(VAC).

Power to the VRS 002 G2 may be supplied from one of the following sources:

- The AC/DC adaptor supplied with the product, 110-230VAC, IEC 60945 nominal.
- DC from an external source, 12-24VDC, IEC 60945 nominal.
- From the RDI or RVI port on the DM100 VDR, if not used for other equipment.

3.4 Description of the front panel



Power connectors:
Power must be supplied <u>either</u> via the barrel connector on the far left or via the Wago connector pin 1 and 2.

Serial interface:
The serial interface on the VRS 002 G2 (Wago pin 3-8) is currently not used – do not connect anything here.

Digital input:
Is currently not used.

Reset:
A paper clip or similar may be used to bring the VRS back to a known state see section 3.5.

Vertical slide switch:
The switch is used to switch between normal operation and extended access. Some functionality will be disabled in normal operation mode; for example, remote initiated system reboot and access to the VDR's configuration interface.

LAN1 – LAN5:
See section 4.3.

USB 3.0 port
Is currently not used

STATUS LED:
See section 3.6.

3.5 Description of the reset switch

A reset switch behind the front panel may be activated using a paper clip or similar.

3.5.1 Reset of address for configuration interface

Activate the reset switch for two seconds until the STATUS LED blinks two times. Release the reset switch and wait until the STATUS LED starts blinking (0.5Hz).

While in this mode, VRS's configuration is accessible via LAN 1 using URL https://10.1.0.80.

3.5.2 Reset to factory settings

Activate the reset switch for seven seconds until the STATUS LED starts blinking at 2Hz. The VRS will then reset the configuration to the factory setting and restart.

3.6 Description of the STATUS LED

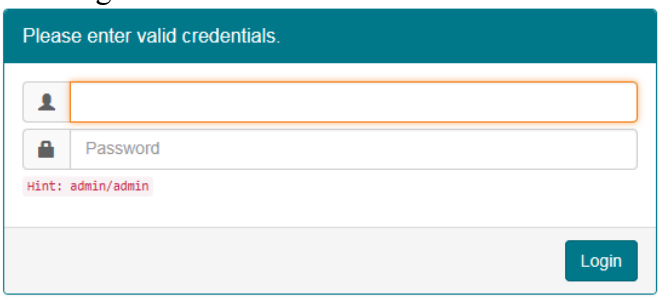
Red:	Error condition or not booted
Yellow:	System is booting
Green:	Normal operation

4 Configuration

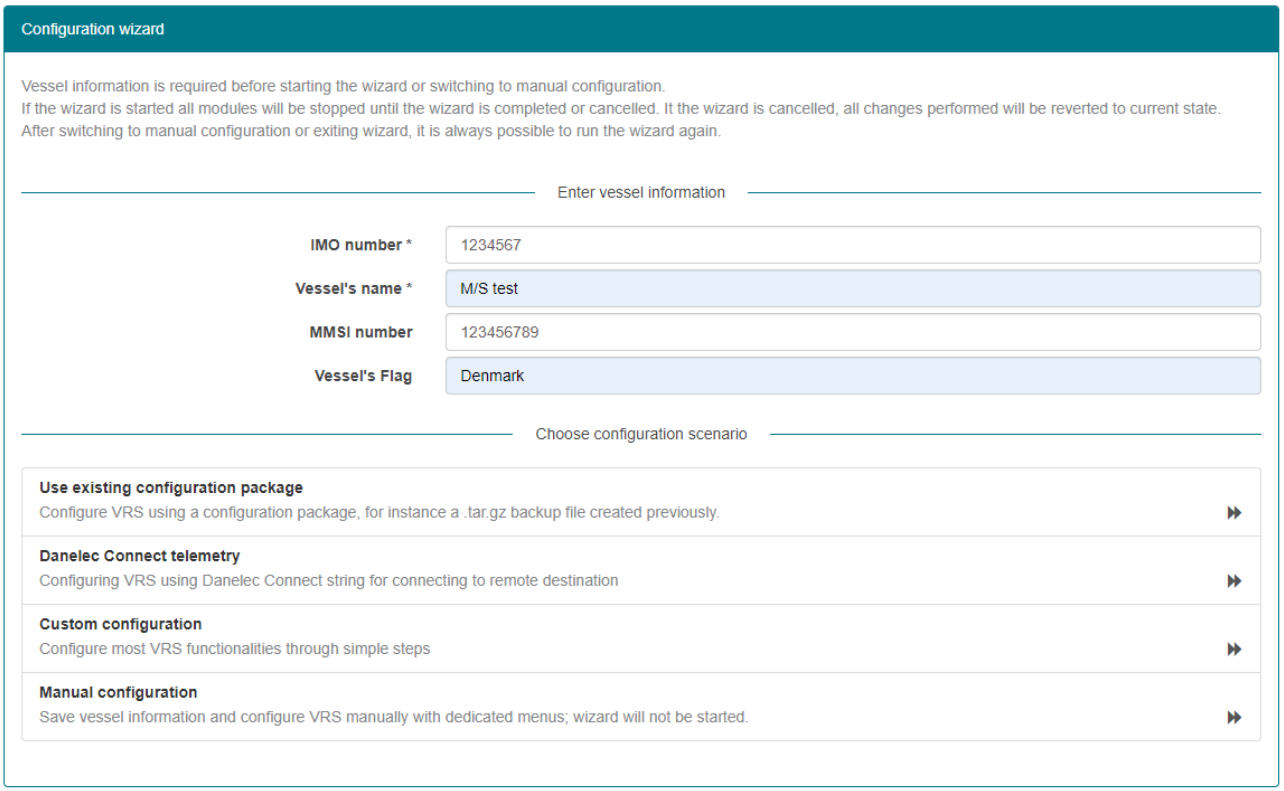
The VRS 002 G2 must be configured from a PC with an Ethernet Interface using an Internet browser.
Connect the PC to LAN 1 on the VRS 002 G2 using a standard RJ45-RJ45 patch cable.

The default URL for the configuration interface for the VRS is https://10.1.0.80. When attempting to connect to the VRS, the browser may display a warning about the security certificate of the VRS. This is due to the fact that an IP address is used rather than a name. The warning must be ignored.

The default password and username are admin/admin. The default password and username should be changed see section 11.1.

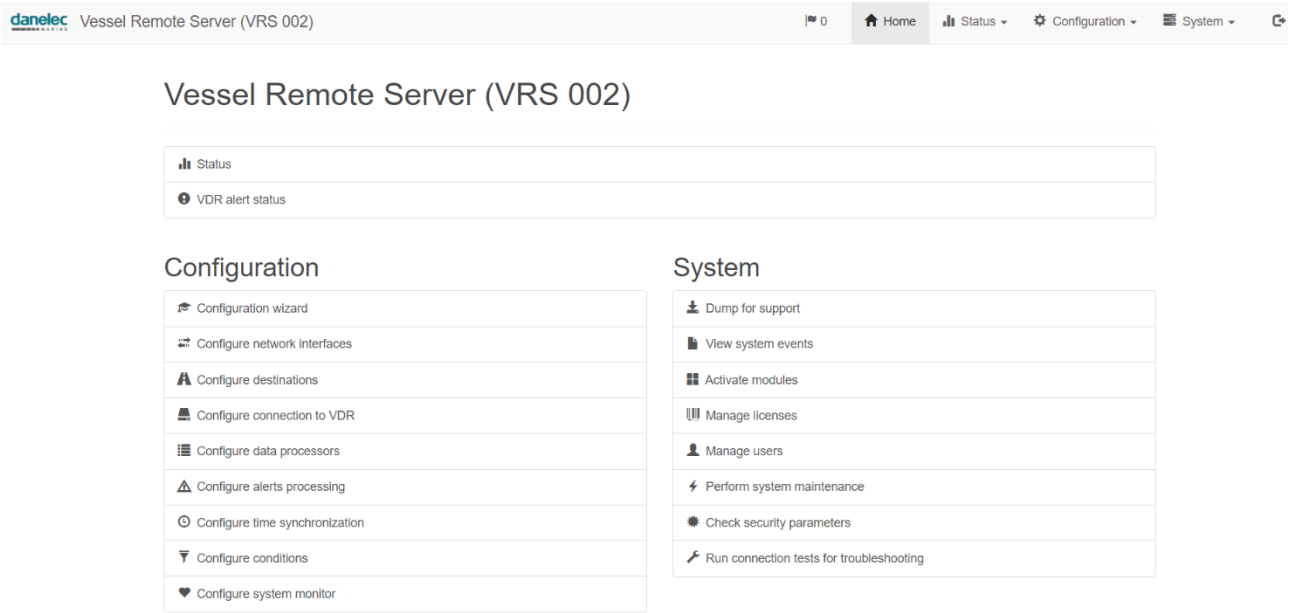


4.1 Configuration wizard



If the VRS is unconfigured, the above page will be displayed. Some basic information about the vessel must be entered, most noticeable the IMO number. If the vessel has no IMO number assigned enter something else which may be used for identifying the vessel, but avoid entering a seven-digit number which is the format for an IMO number.

4.2 VRS 002 G2 home (manual configuration)



The VRS 002 G2 home page links to all major configuration and status pages.

4.3 Configuration of network interfaces

It is possible to configure up to five independent network interfaces on the VRS 002 G2. Note that the VRS will never route or switch between the LAN ports.

4.3.1 VRS standard configuration

To facilitate easy service and maintenance of the installed systems, the LAN ports must be allocated (when possible) as described below:

LAN port 1: Configuration interface
LAN port 2: VDR interface or network interface for mini VDR
LAN port 3-5: For LAN and WAN access

4.3.2 Configuration of connection to VDR

(For installation where the VRS serves as a mini VDR, see section 6)
Network interface 2 using LAN port 2 is pre-configured for connection to the VDR.
If changes are necessary, go to "Configuration->Network->Network Interface 2" and "Configuration->VDR connection".

The status of the VDR connection may be checked by going to “Status->Dashboard”.

VDR live connection

State	Last update	Information	Alarms	Serial	Analog/Digital	Audio	Image
✔	09/02/2017 14:54:12	OK	21	2878	0	0	0

4.3.2.1 Vessel information

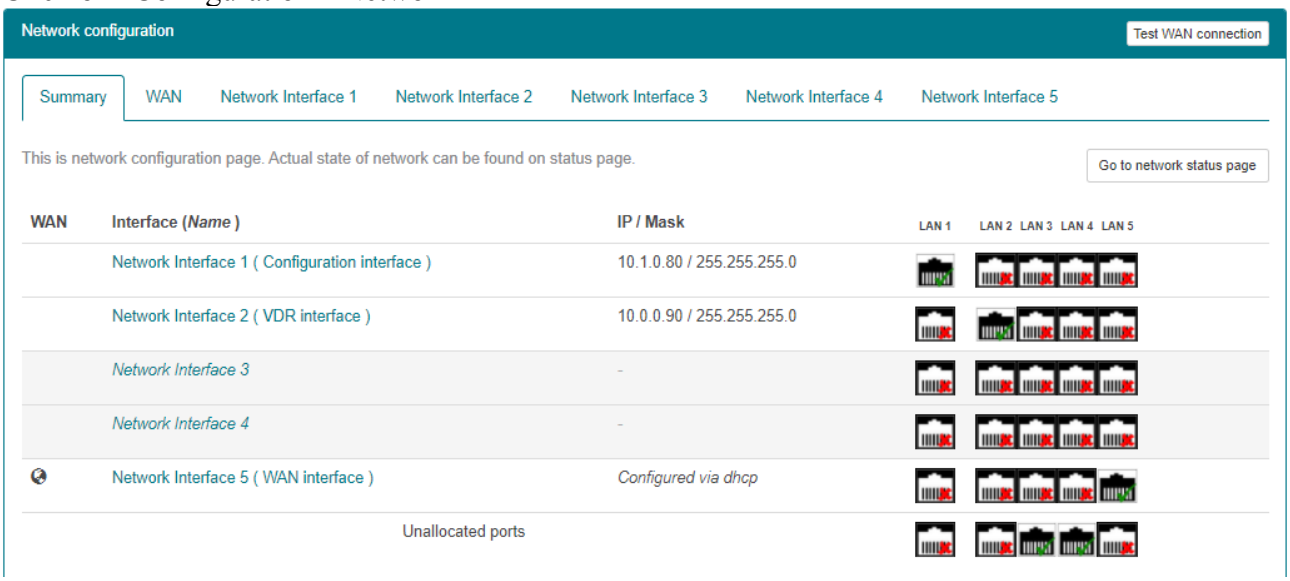
The information about the vessel should be updated automatically if the VDR is configured correctly. Go to “Status->Vessel” and check this.

Dashboard | Vessel | System | Network | Data Processor Reports

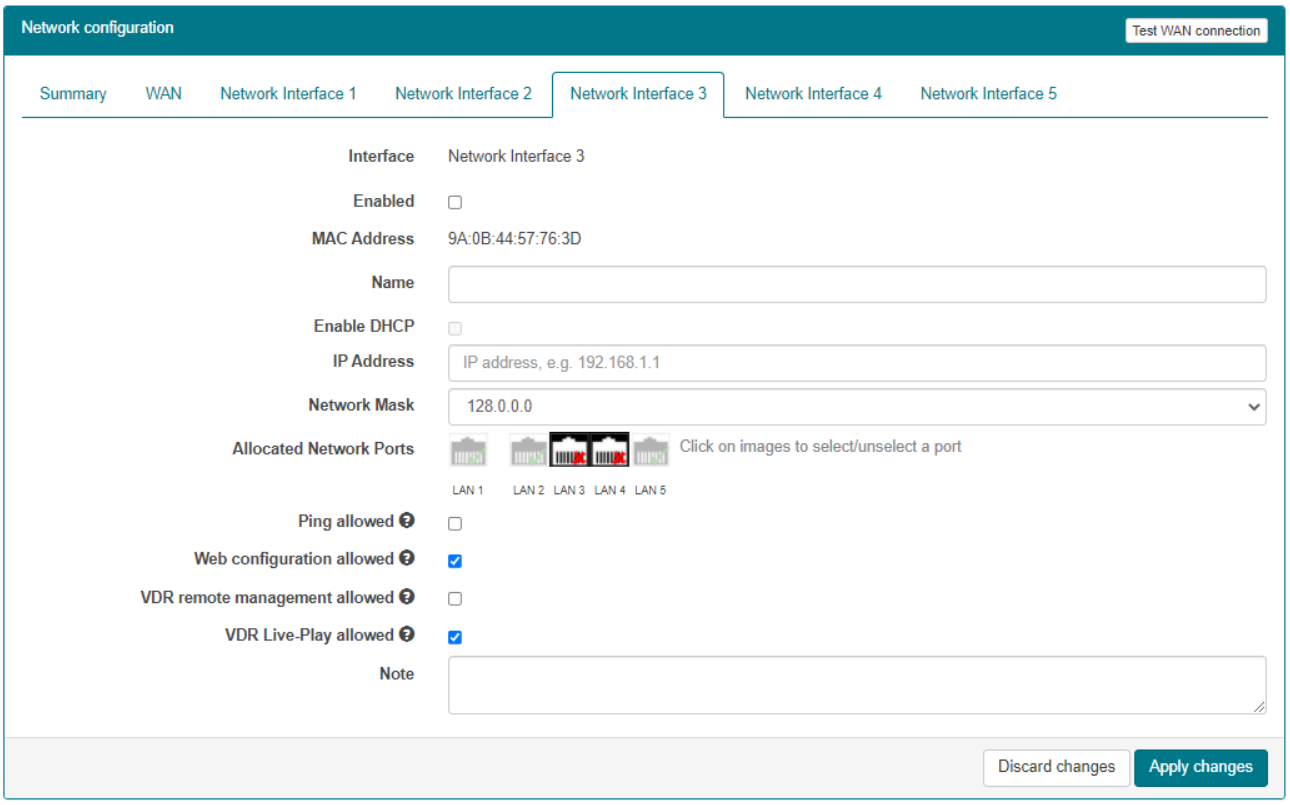
IMO number	123456
Vessel's name	M/S test
MMSI number	123456
Vessel's flag	Denmark

4.3.3 Configuration of connection to LAN

Click on "Configuration->Network"



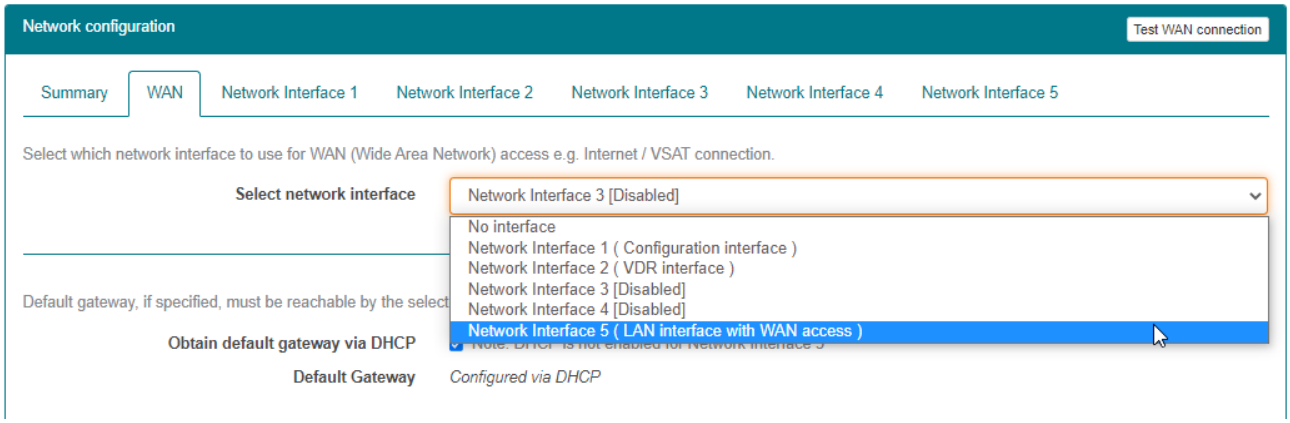
Click on “Network Interface 3” to set up a new network interface



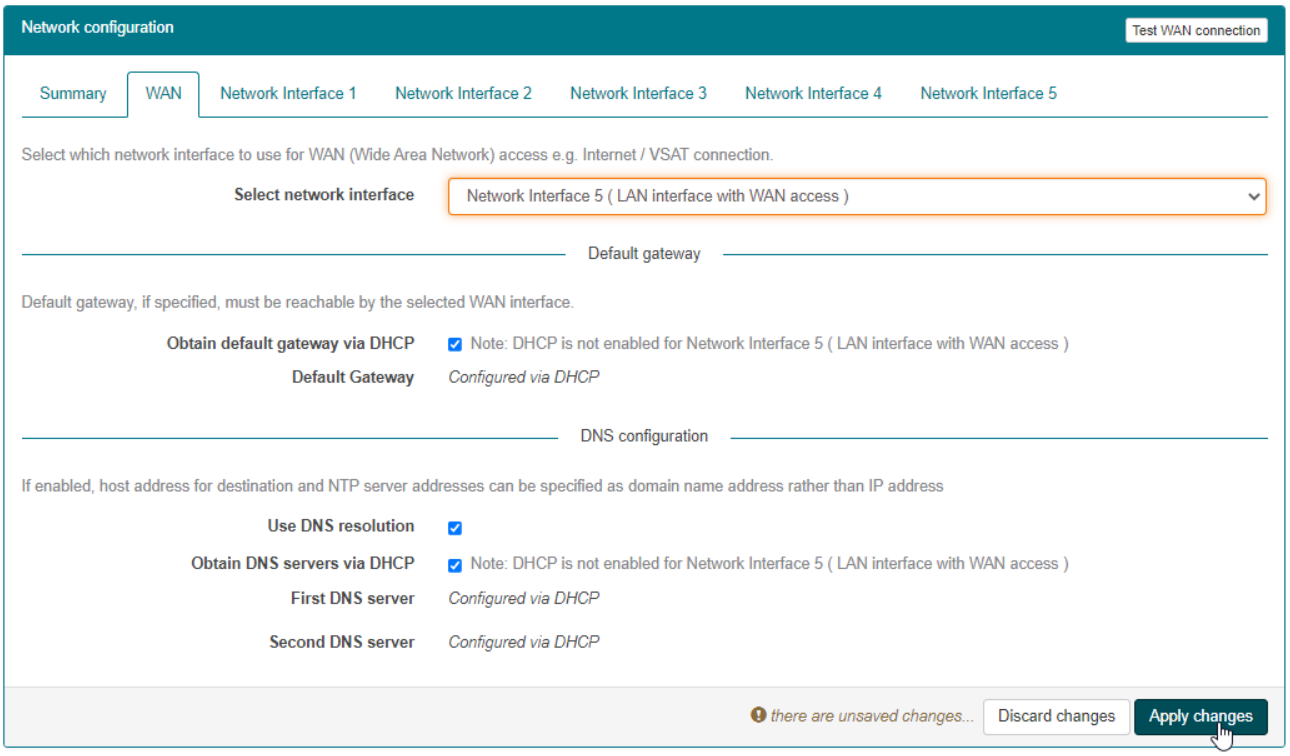
Fill in relevant fields and allocate a minimum of one LAN port for the network interface.
Click on “Apply changes”.

4.3.4 Configuration of the default gateway (WAN access)

Click on “Configuration->Network->WAN”.



Select which network interface to use for WAN access



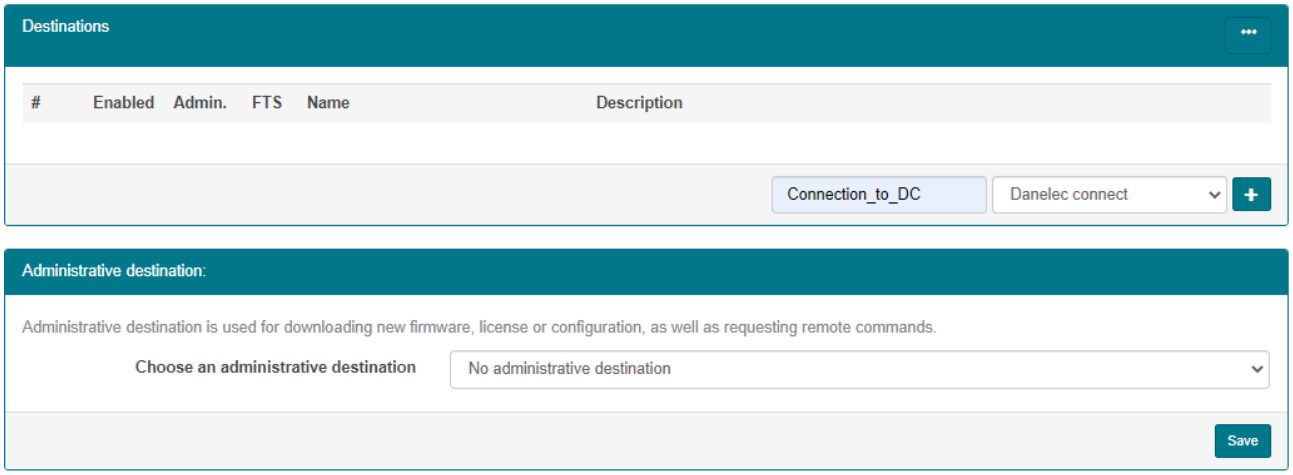
Enter the IP address of the default gateway and other relevant parameters and click on “Apply changes”.

4.3.5 DHCP support

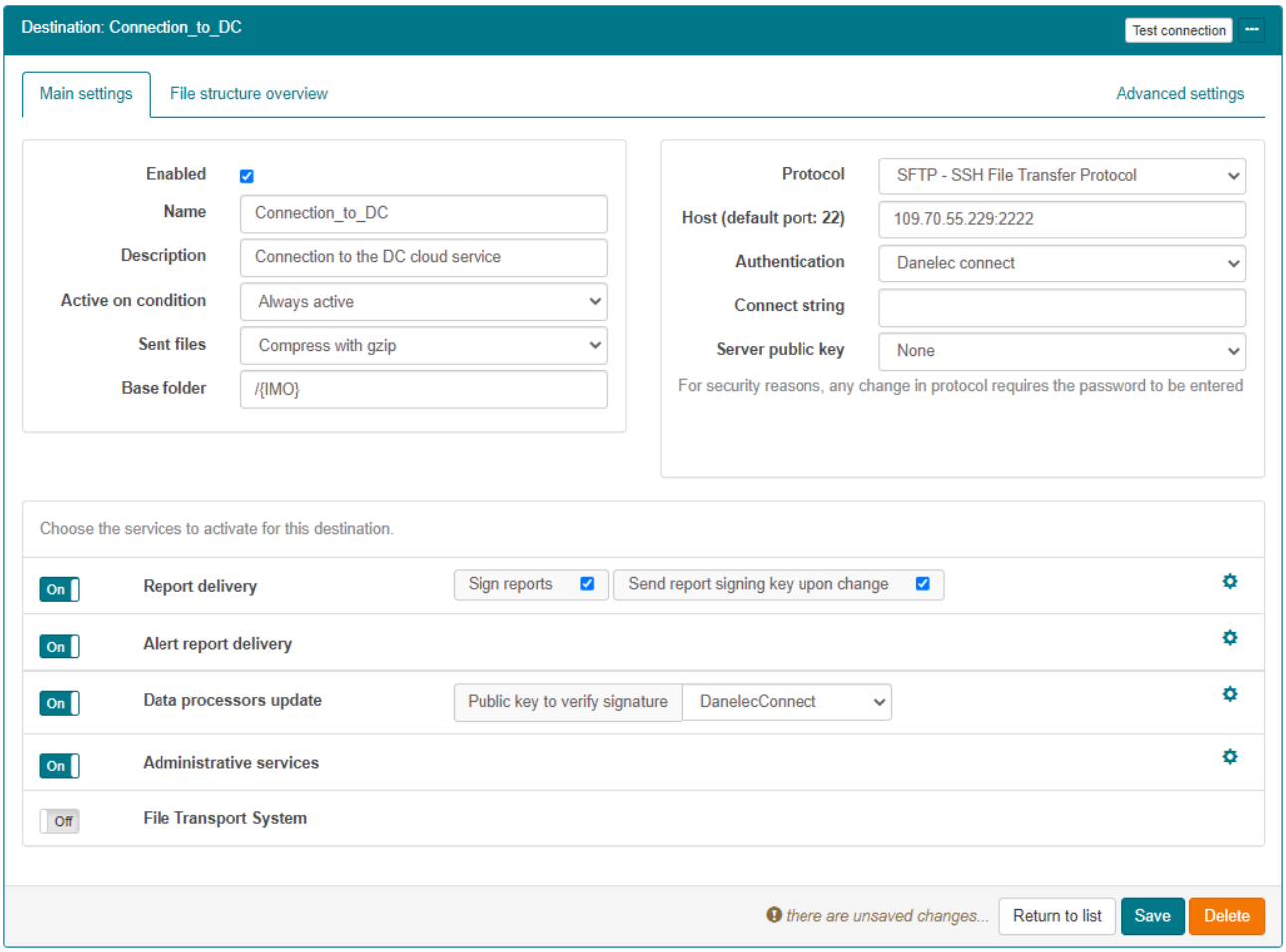
For one Network interface, DHCP may be used for assigning the IP address for the interface and optionally the addresses for the default gateway and DNS server. The assigned IP address will be ignored, and the interface will remain inactive if the address conflicts with the address allocation for other network interfaces.

4.4 Configuration of destinations

Click on “Configuration->Destinations”.
This configuration page is related to the transmission of telemetry data. A destination in this context is the IP address and other information for a destination server.



Enter a name (this will become the destination name, which is the ID for the destination) and click on “+”.



Enter the necessary parameters. The connection may be tested by clicking on "Test connection". Click on "Save". Note that the layout of this form depends on the chosen protocol and template (see the previous page).

4.4.1 Description of parameters

Enabled: A destination may be enabled or disabled. When disabled, all files generated by the data processors for the destination will be flushed.

Name: A unique ID and reference to the destination used by the data processor(s).

Description: Free text.

Active on condition:
See section 4.5.

Send files (compression): Transmitted data (reports) may be compressed.

Base folder: It is possible to organize the files on the server in different ways. Typically the vessel's IMO number is used as a name for the "root" folder, but this is configurable; see section 4.4.2.1.

Protocol: Protocol to be used for data transfer.
Three options exist:

- FTP and SFTP: The VRS will send .xml formatted files (reports) to the server.
- HTTP endpoint: The VRS will act as a server providing a REST service that client(s) must poll. The format for the reports, in this case, is JSON (not possible for DanelecConnect).

For information about file formats, contact Danelec Marine Support.

Host address: The IP address or DNS name of the target server/system.

Authentication: Authentication method to be used.

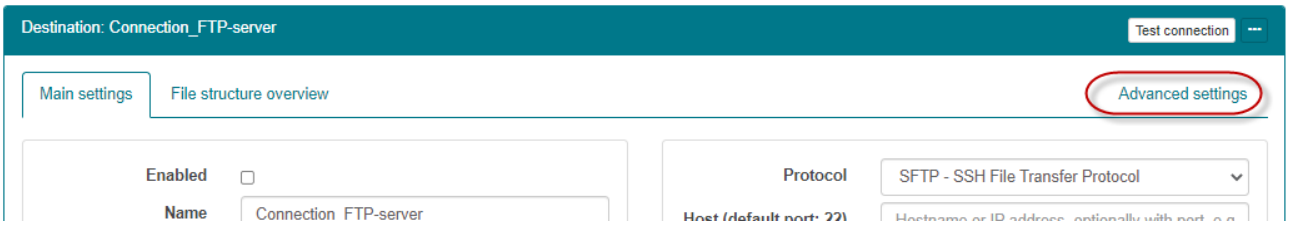
Connect string: The connect string is a token that will allow access to the server. Contact Danelec Marine Support for obtaining a connect string.

Username: The user name for the target server/system – not relevant for DanelecConnect.

Password: Password for target server/system – not relevant for DanelecConnect

List of services: What the destination may be used for must be configured – tick one or more checkboxes.

4.4.2 Description of the directory structure and file name for reports



The directory structure for the reports may be user-defined in the under advanced settings for a destination. The directory path and filename for a report may be a combination of free text, fields updated by the VRS, and text generated by the VRS. Do not change this for a destination to the DanelecConnect cloud service.

An example of this is shown below:

IMO number: 1239999
Report name: SOG_COG_report
Date for report generation: 2016-11-28
Time for report generation: 08:22:15

"Report final file" is: /{IMO}/{REPORTFILENAME}

The VRS will replace {IMO} with 1239999 since "IMO" is on the list of strings that the VRS must replace.

"REPORTFILENAME" is also on the list, but is this string will be replaced with:
< Report name>-<YYYYMMDD>-<hhmmss>.xml

In this case, the path and name for the file will become:
/1239999/SOG_COG_report-20161128-082215.xml
If we want to gather reports of this type in a separate folder called "Short_reports" and we want the vessel's IMO number to be part of the filename, the following "definition" may be used:

/Short_reports/IMO{IMO}_{REPORTFILENAME}

In this case, the path and name for the file will become:
/Short_reports/IMO1239999_SOG_COG_report-20161128-082215.xml

4.4.2.1 List of strings the VRS will replace

Strings to be replaced must be framed by two curly brackets, i.e. {}.

String	Replaced by
IMO	Replaced by the configured value for the vessels IMO number see section 4.3.2.1 and 6.4
VESSELNAME	Replaced by the configured value for the vessel's name, see section 4.3.2.1 and 6.4
DEVICEID	Each VRS has a unique H/W ID. This ID may be used for the path or the filename. However, if the VRS hardware fails and is replaced, the ID will change.
DATAPROCESSORID	The name for the data processor which has made the report
REPORT FILENAME	<reportname>-<timestamp>.xml Where reportname is the name of the report configuration and time stamp is: YYYYMMDD-hhmmss
REPORTNAME	The name of the report configuration associated with this report
TIMESTAMP	Replaced by YYYYMMDD-hhmmss
DATE	Replaced by YYYYMMDD
BASE	Replaced by the name of the base folder see section 4.4.1

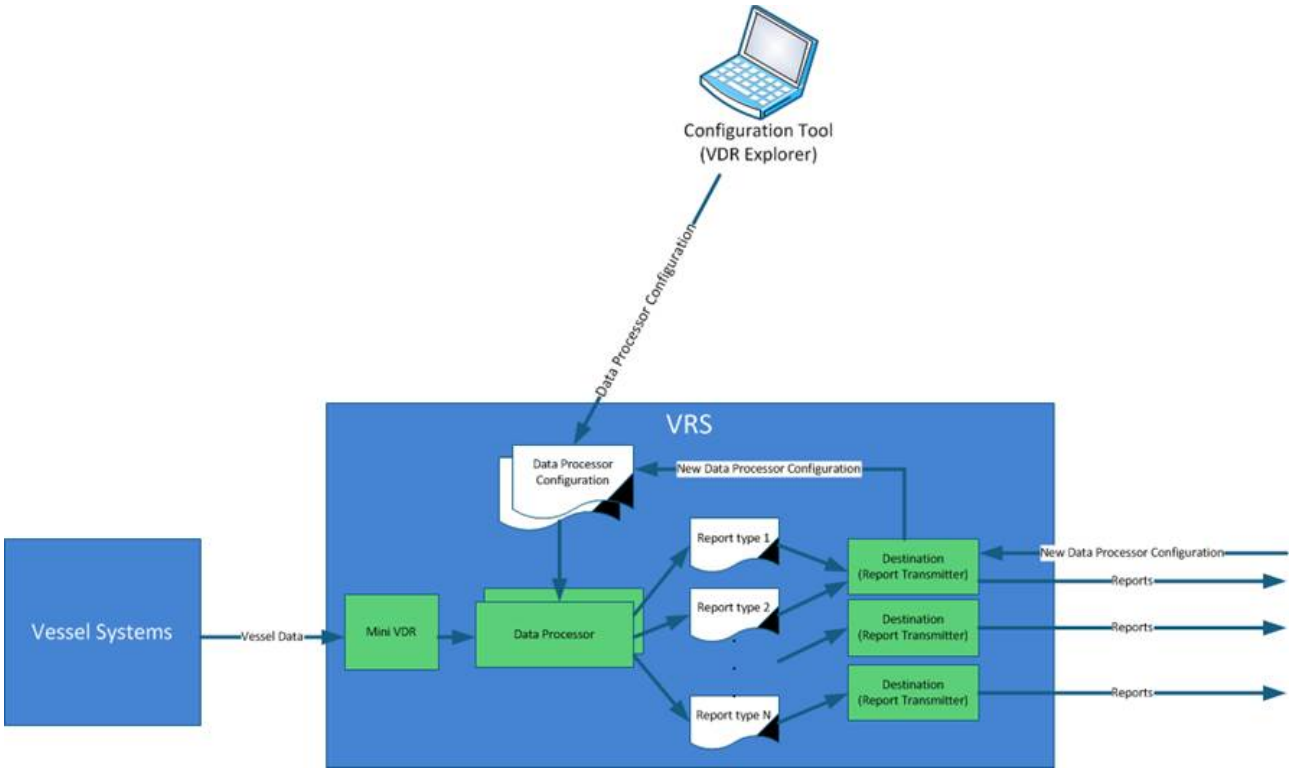
The principles described in this section are also applicable for the directory structure for remote configuration upgrade of the VRS; see section 7.

4.5 Configuration of (transmit) conditions

Go to Configuration->Conditions
The VRS may be configured to halt the transmission of reports until a specific condition is met. Currently, only one type of condition is supported based on whether it is possible to successfully "ping" a specific server.

5 Configuration of data processors (report generation)

5.1 Overview



5.1.1 Data processor

Reports are generated in the VRS by data processors. The current VRS supports a maximum of two data processors. A data processor configuration is created using the VDR Explorer tool; i.e. the VDR Explorer is used for creating a configuration file that is uploaded to the VRS.

5.1.2 Report configuration

A data processor configuration contains the required information for generating one or more types of reports. The content of the reports, the destination for the reports, and how often they are sent may be configured individually for each type of report.

5.1.3 Destination

The possible destinations for reports (typically the IP address of a server) must be configured in the VRS in advance (see section 4.4). A destination name must be defined for each destination. A report configuration must contain a reference to a destination. It is currently not possible to download the list of destinations from the VRS to the VDR Explorer, so the VDR Explorer list must be populated manually (see section 5.2.2).

5.2 Configuration

The VDR Explorer is used for creating a data processor configuration. Suppose a VDR Explorer configuration for a vessel exists. In that case, it is easy to create a data processor configuration since the input signals must already be decoded and calibrated due to the requirements for OPT.

In the following description, it is assumed that no VDR Explorer configuration exists, which is unusual.

Remember to set the VDR Explorer to configuration mode before any configuration changes are possible (Mode->Configuration Mode – ON).

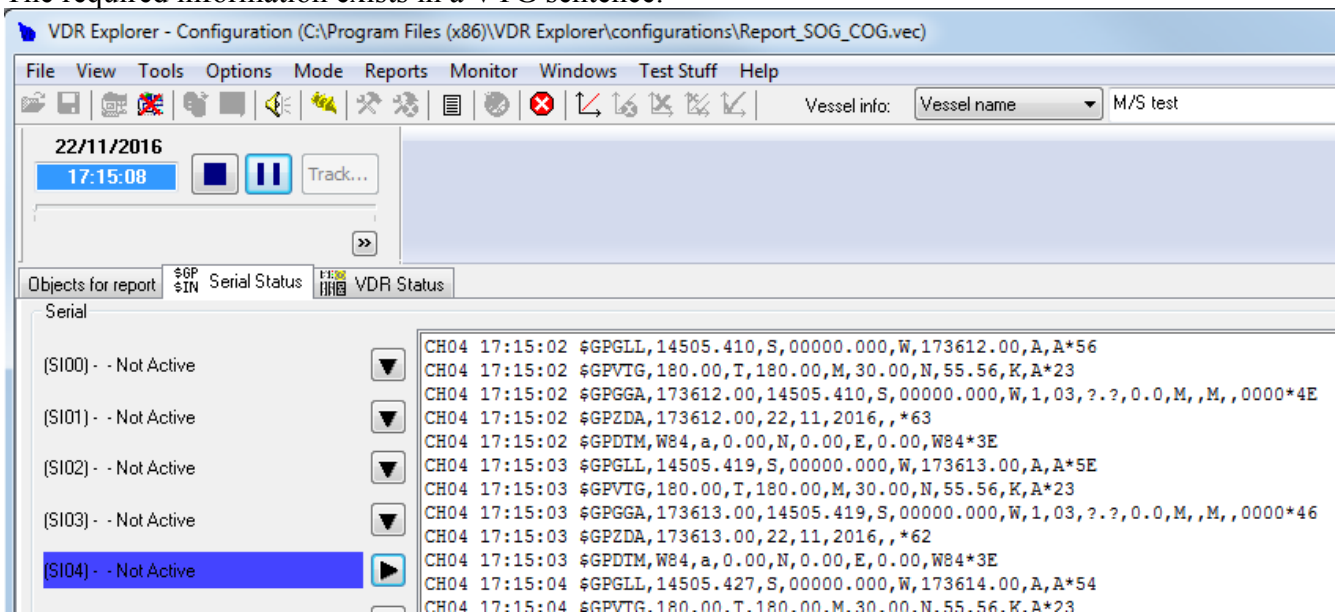
The following steps must be carried out:

Step	Description
Creation of VDR Explorer configuration	5.2.1
Updating the list of destination	5.2.2
Creating a data processor configuration	5.2.3
Upload of configuration to VRS	5.3

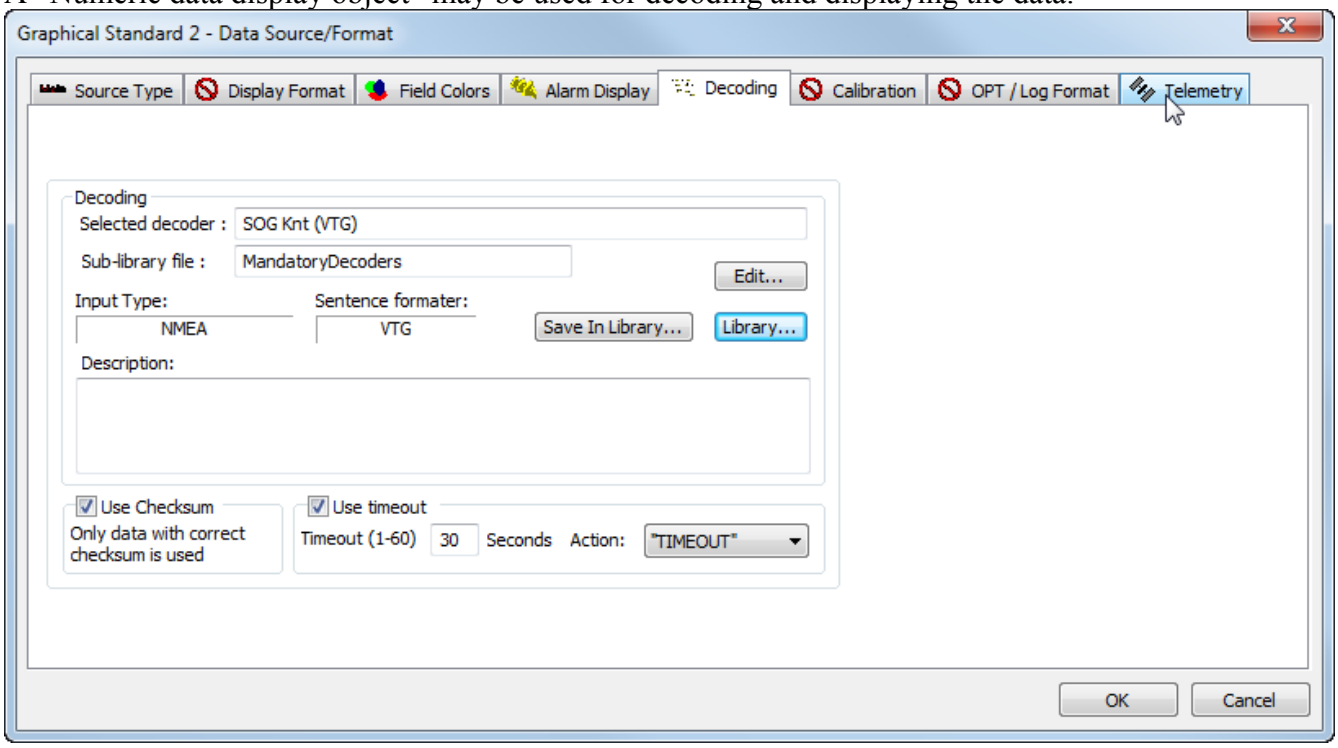
5.2.1 Creation of VDR Explorer configuration

In the example below, a report containing SOG and COG must be generated.

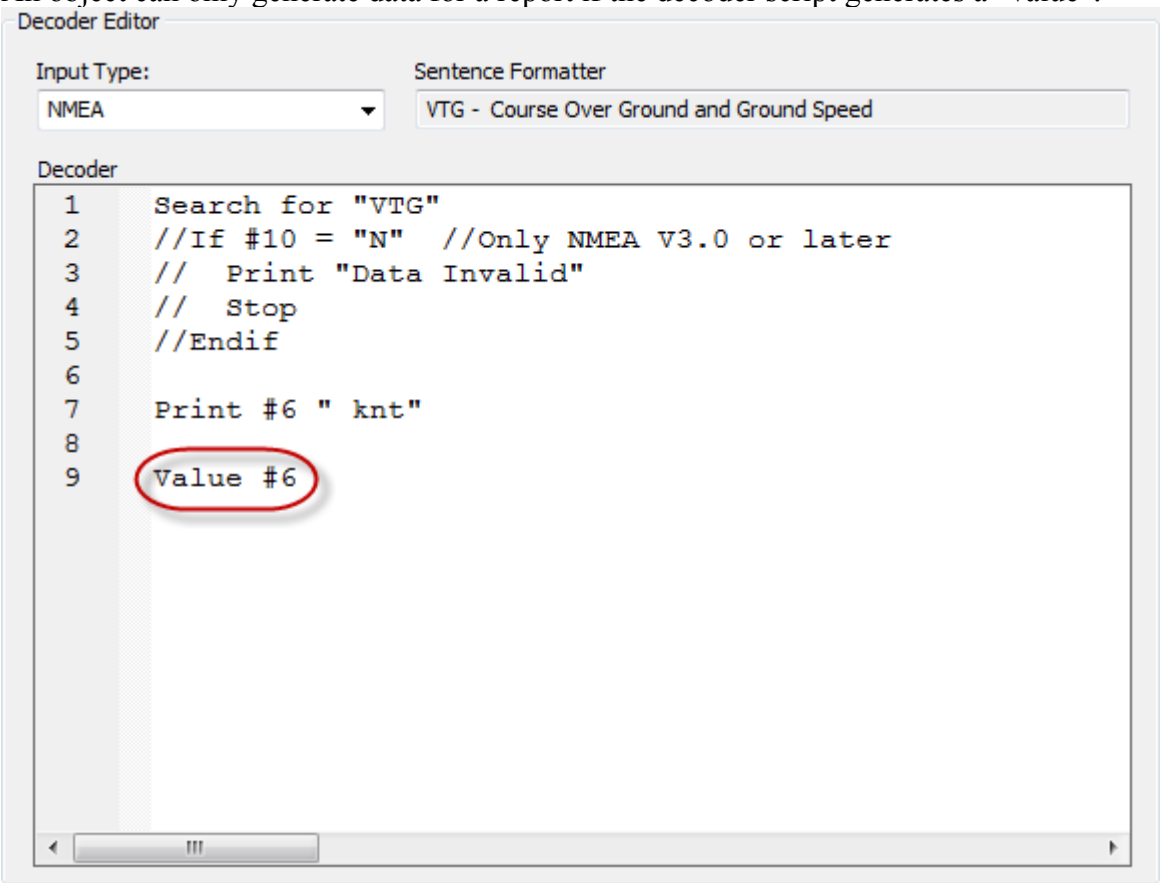
The required information exists in a VTG sentence.



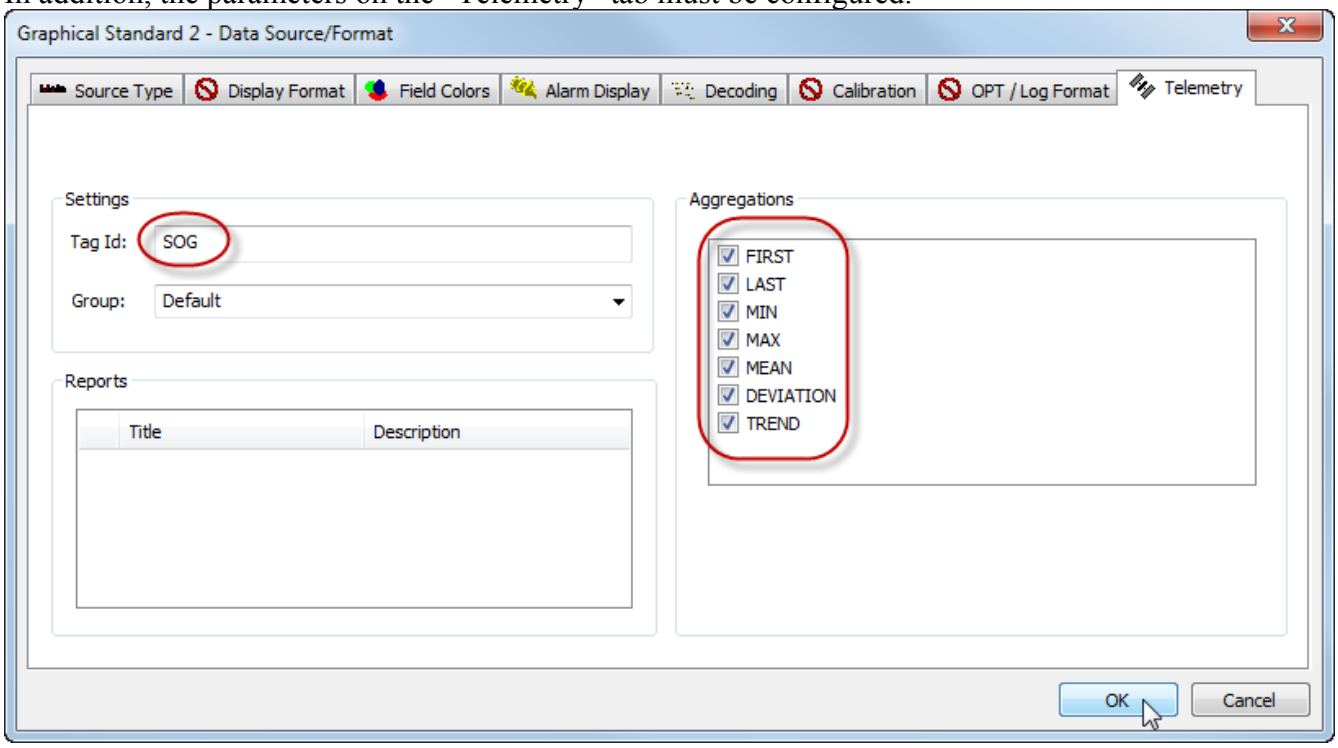
A “Numeric data display object” may be used for decoding and displaying the data.



An object can only generate data for a report if the decoder script generates a “value”.

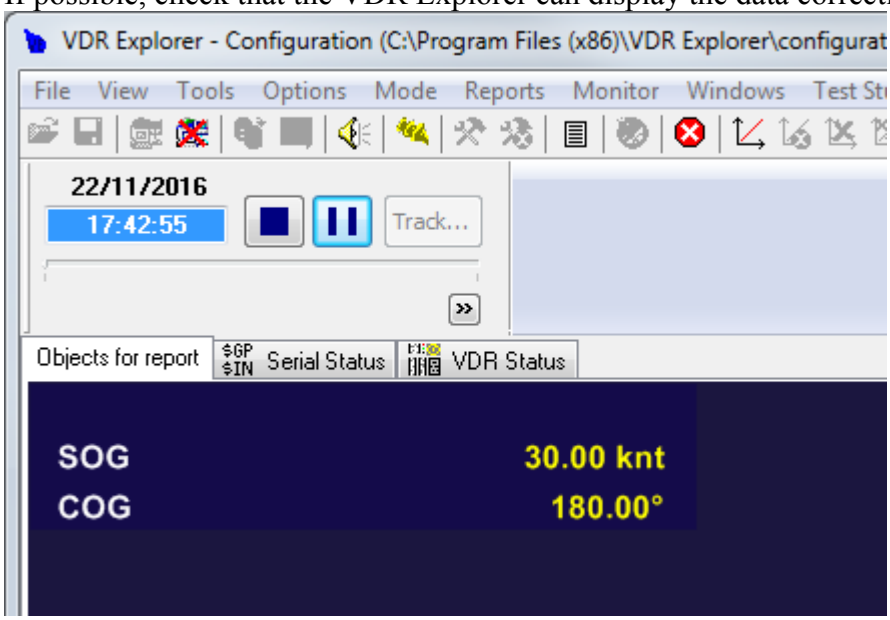


In addition, the parameters on the “Telemetry” tab must be configured.



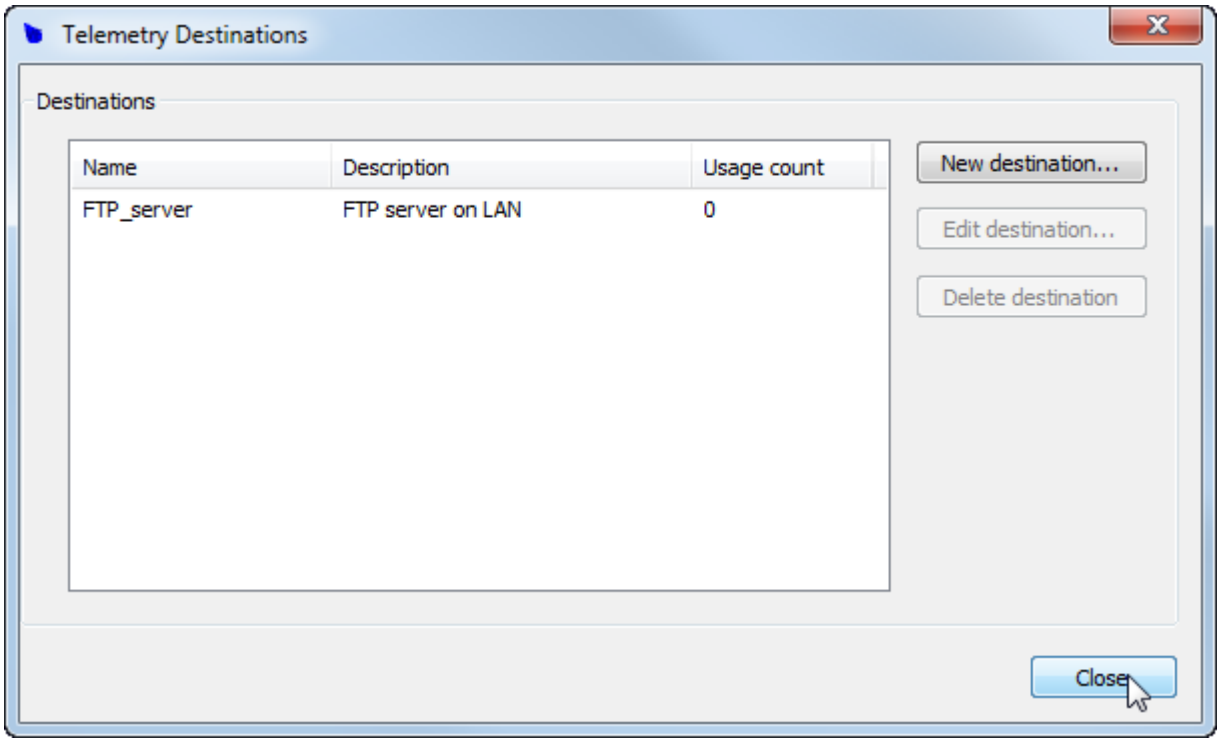
The above steps must be repeated for COG.

If possible, check that the VDR Explorer can display the data correctly.



5.2.2 Updating the list of destination

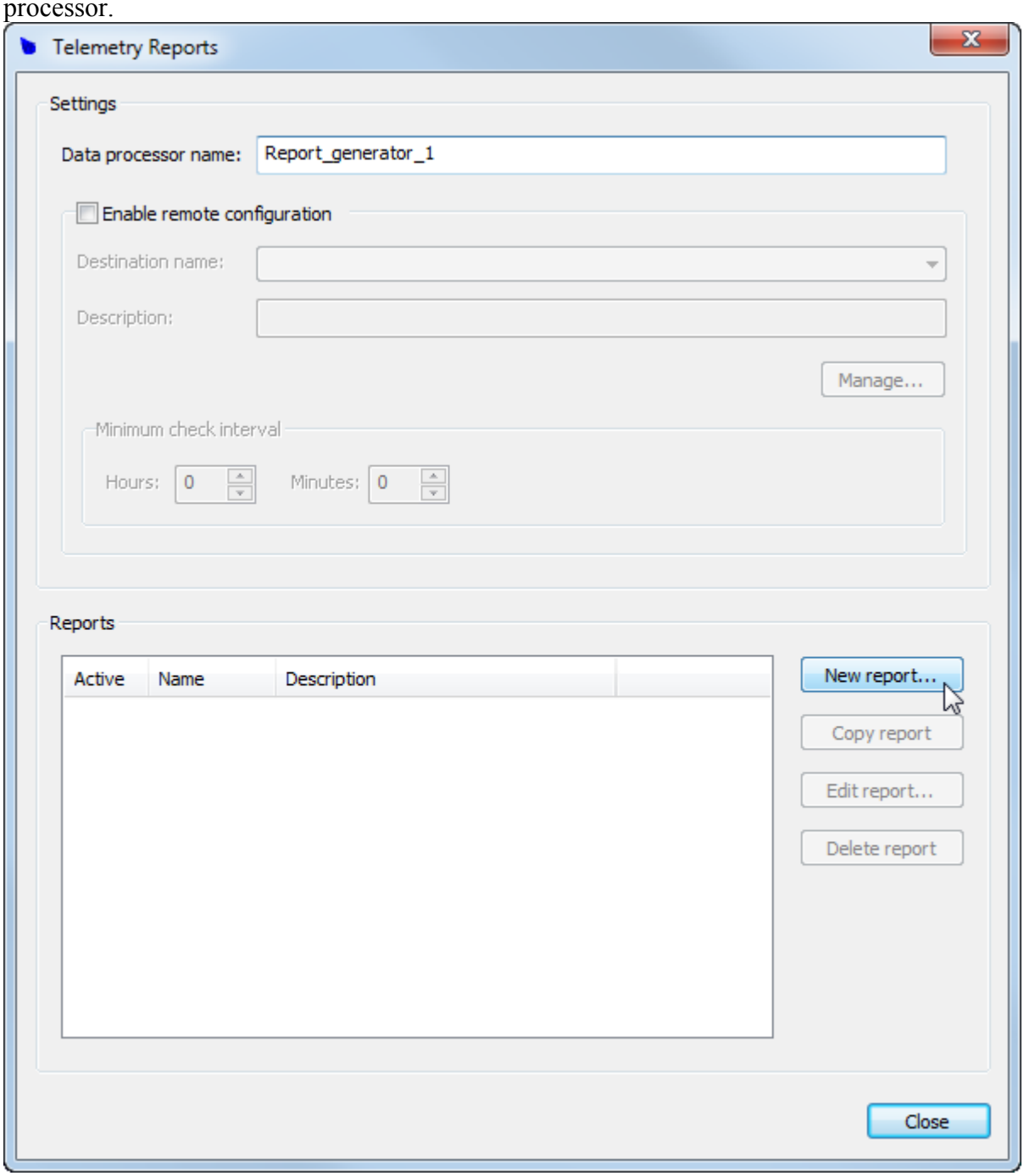
In the VDR Explorer, go to “Reports -> Destination” and update the list of destinations.



It is essential that the name for the destination in the list is identical to the name for the destination in the VRS; see section 4.4 and 5.1.3.

5.2.3 Creating a data processor configuration

In the VDR Explorer, go to “Reports -> Reports” and configure the main parameters for a data processor.

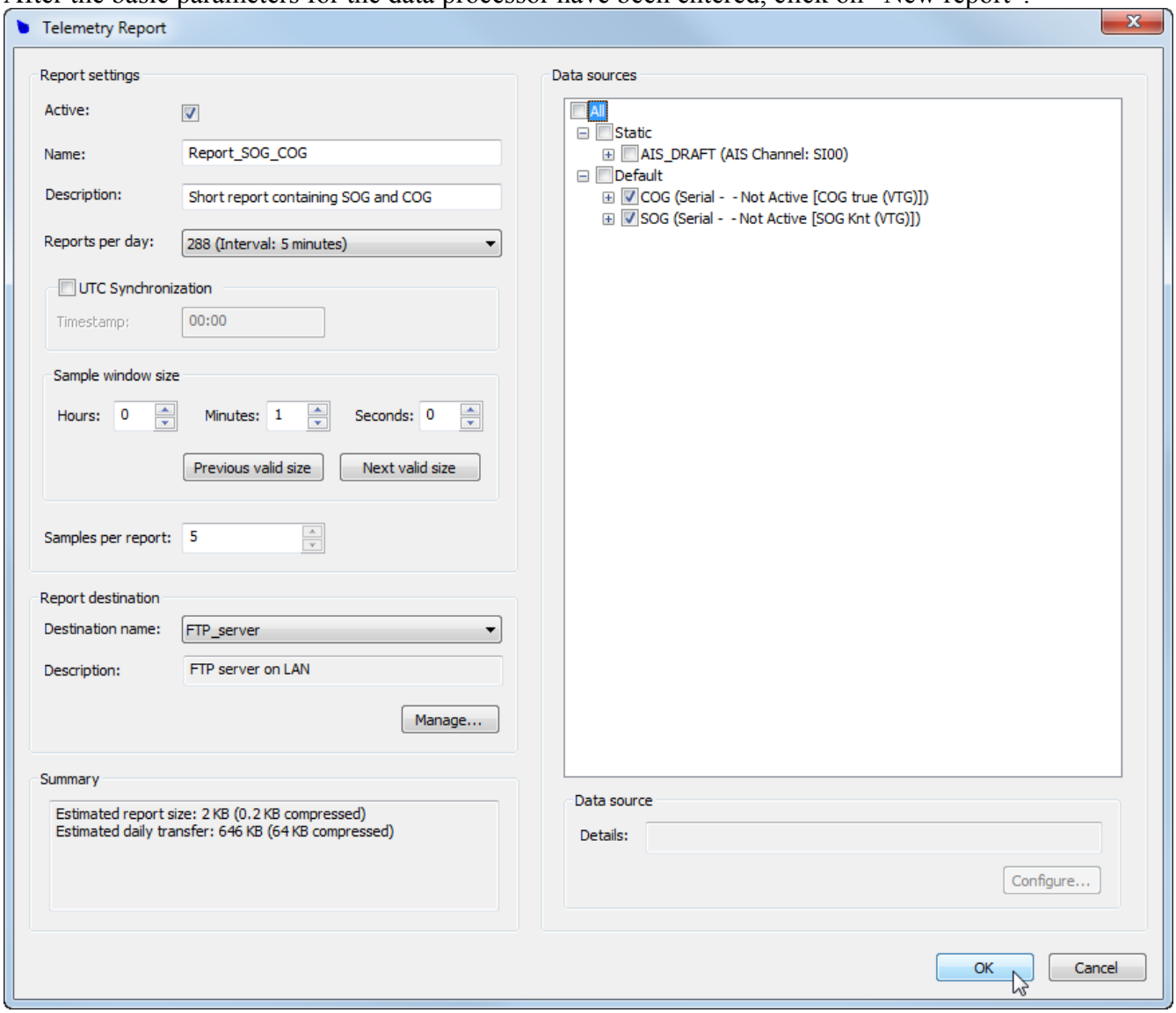


Click on “New report” to create a report configuration.

Data processor name:
The Data processor name will become the name of the data processor in the VRS.

Enable remote configuration:
See section 7

After the basic parameters for the data processor have been entered, click on “New report”.



Click on “OK”

Exit VDR Explorer configuration mode and save the VDR Explorer configuration.

5.2.3.1 Description of parameters

Active:
If “Active” is unchecked, the generation of the report will not become active following the upload of the data processor configuration to the VRS. It is possible later to activate the report generation in the VRS (Configuration->Data Processors).

Name:
The ID of the report. This may become part of the filename for the reports. See section 4.4.2.

Report per day:
This parameter determines the length (time) of the reports.

UTC synchronization:
It is possible to synchronize report generation with, for example, “the top of the hour”. If the report length is 1 hour (24/day) and you want the report 00:00 01:00 …. 23:00 use 00:00 as synchronization timestamp (or any other hh:00).

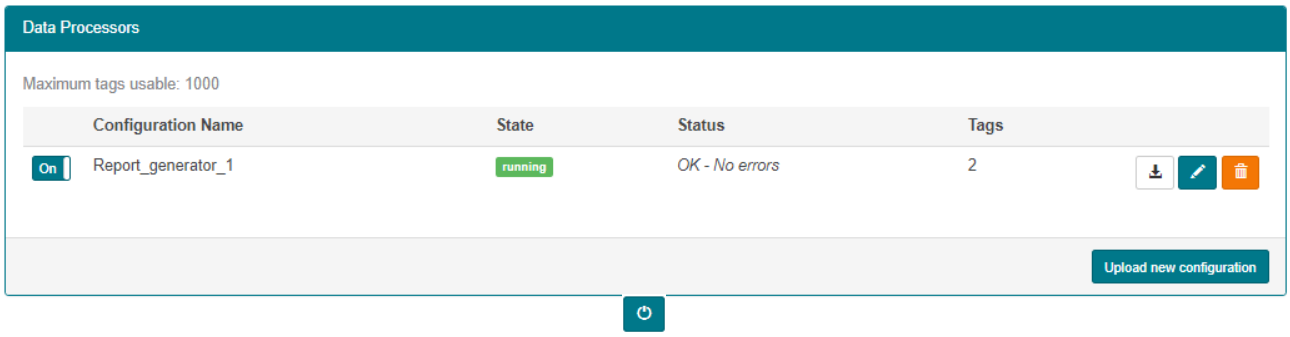
Sample window size:
This parameter determines how often values are added to a report. If the report length is one hour and the sample window size is 10 minutes, each report will contain six sets of values.

Samples per report:
The value for the parameter is typically calculated by the VDR Explorer on the assumption that a continuous stream of data is required. But it is possible, for example, every day, to send a short report with a burst of data using a small sample window size. For some applications, this may be used to verify that the averages related to a larger window size do not hide unexpected fluctuations.

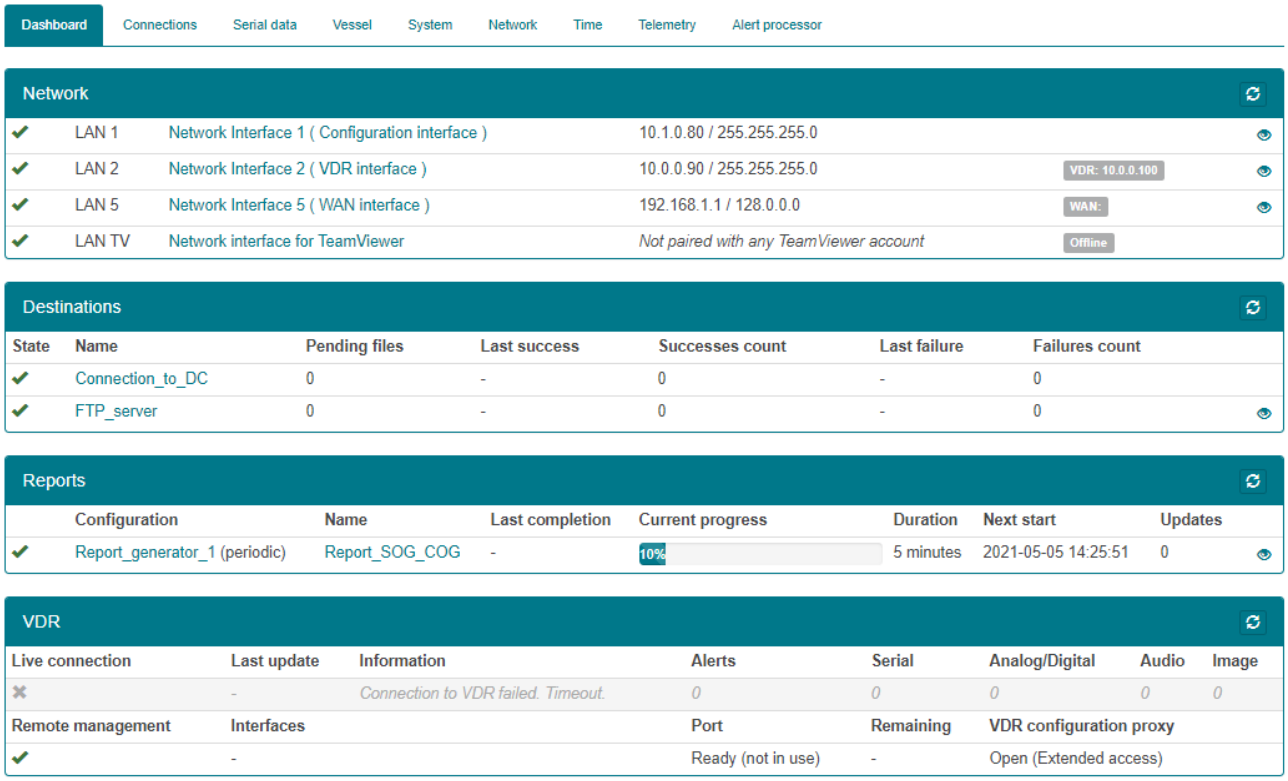
Report destination:
The destination for the report, see section 5.1.3.

5.3 Upload of configuration to VRS

In the VRS user interface, go to “Configuration -> Data Processors” and upload the VDR Explorer configuration to the VRS.



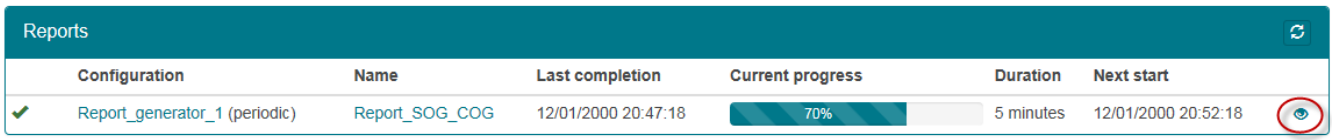
Go to “Status->Dashboard” and check the progress for the report generation.

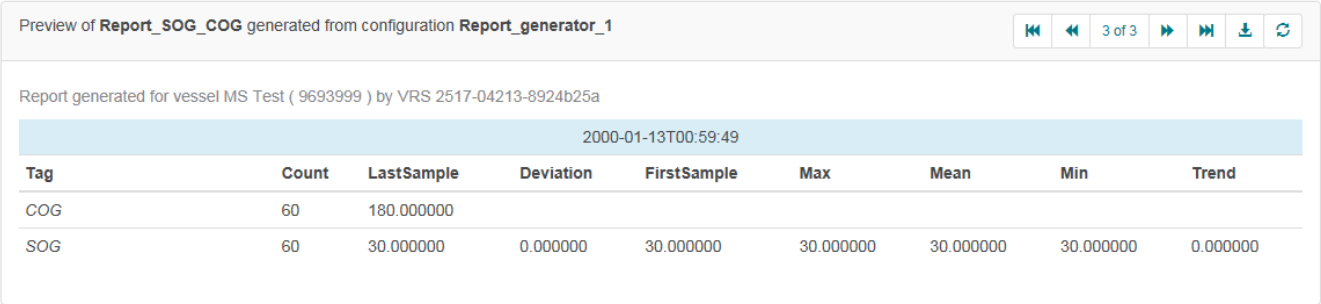


Note that “Successes count” under “Destinations” will be incremented when a report has been delivered successfully to the destination.

5.3.1 Preview of report

It is possible to preview the report currently being generated, click on the eye symbol to the right.





Report preview

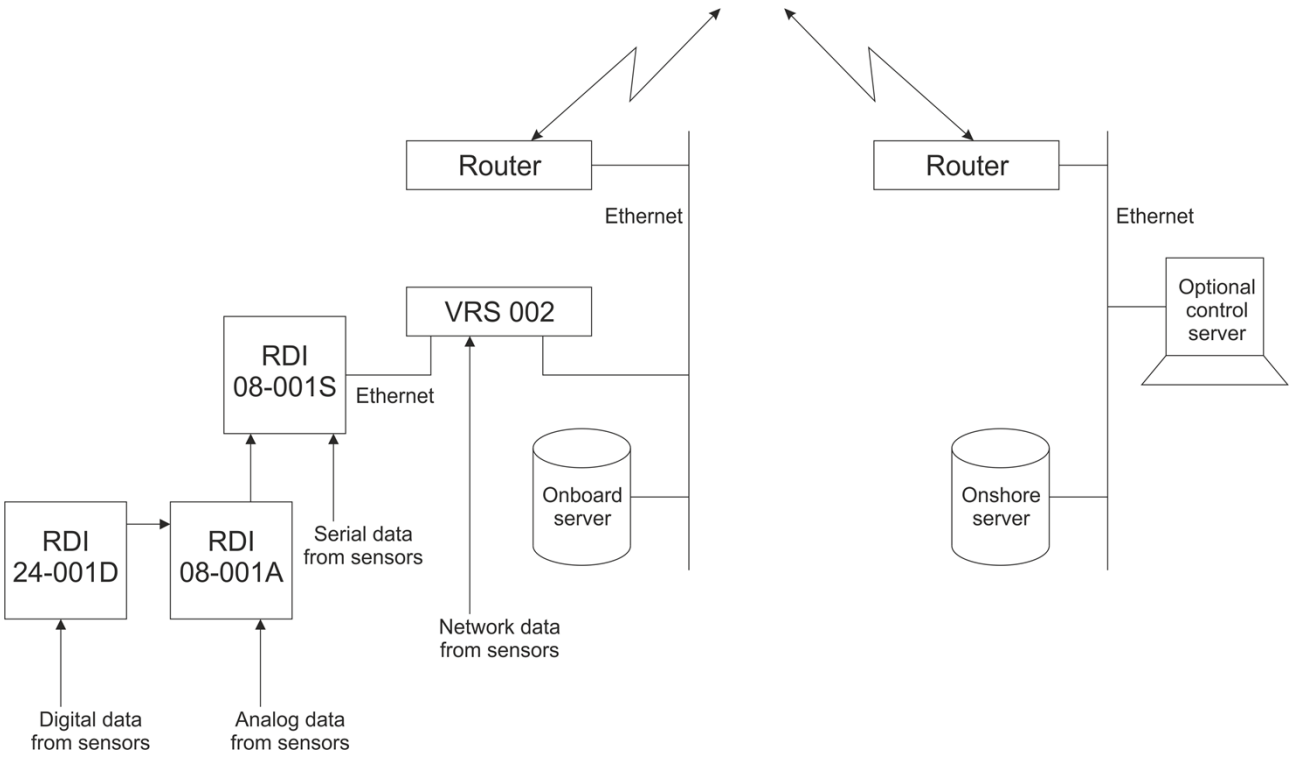
6 Configuration of a mini VDR

For vessels where no Danelec Marine VDR is installed, the VRS 002 G2 may still be used for data collection in conjunction with one or more RDIs.

The following steps must be carried out:

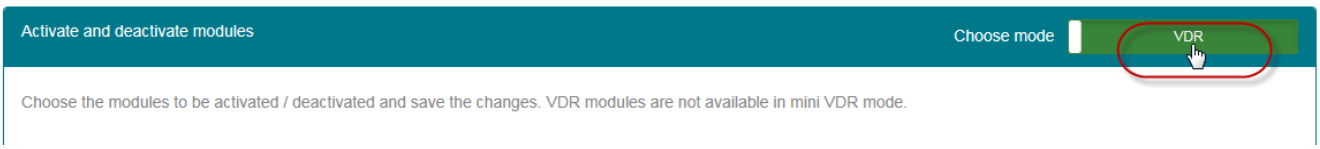
Step	Section
Enabling the mini VDR module	6.2
Configuration of network interface	6.3
Configuration of global parameters for mini VDR	6.4
Configuration of serial interfaces on RDI	6.5

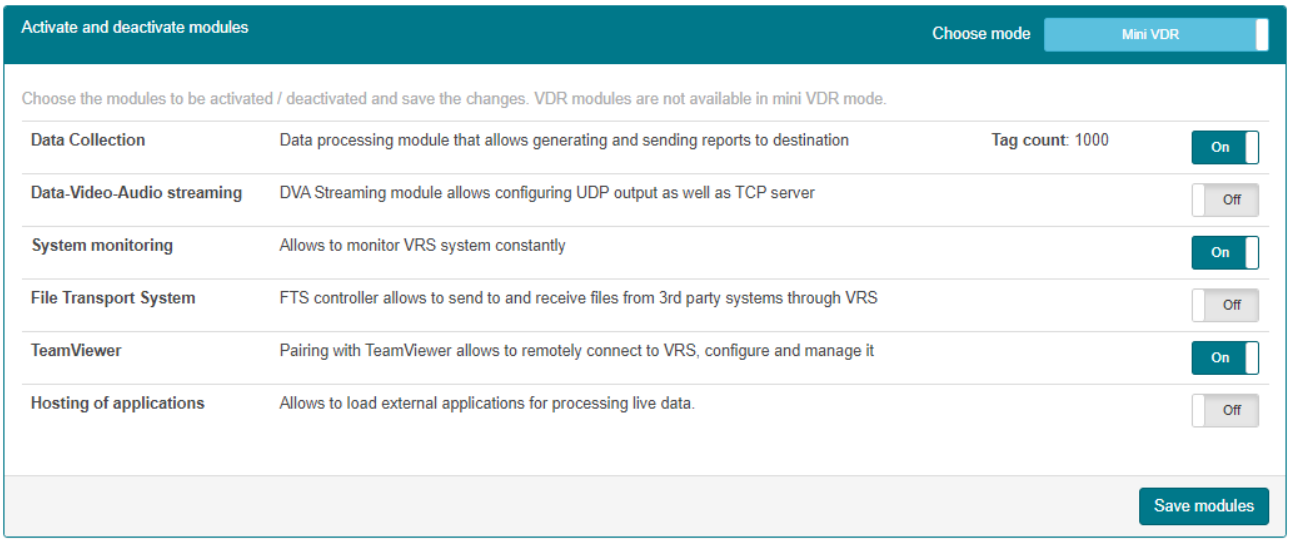
6.1 System overview



6.2 Enabling the mini VDR module

Go to “System->modules” and toggle mode.



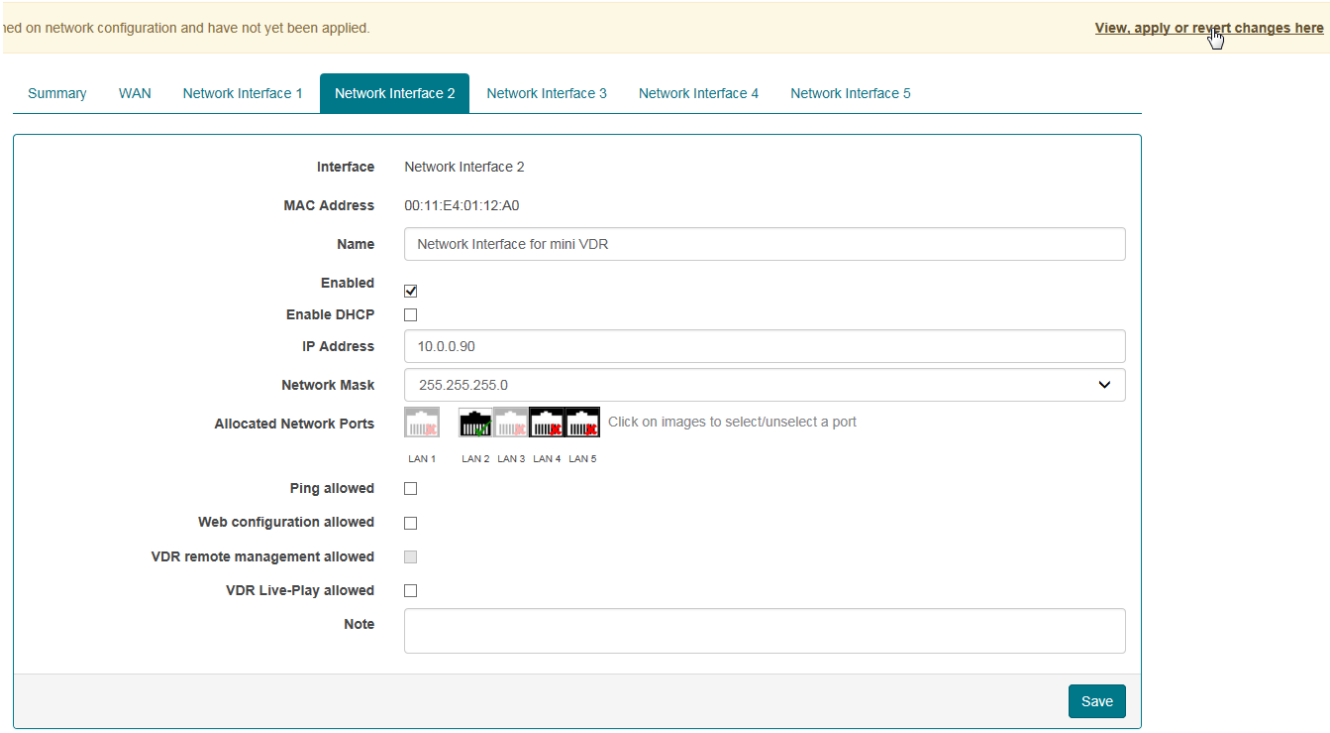


A connection to a VDR and the mini VDR cannot co-exist, so this will disable functionality related to the VDR.

6.3 Configuration of the network interface for mini VDR

Go to “Configuration->Network” and create a network interface for the mini VDR.

The IP address/Network mask must be set to an address that is not in conflict with the other network interfaces and outside address range 192.168.238.0/255.255.255.0, which is reserved for the RDIs. The configured IP address/Network mask may be used as the destination address for network data sent to the mini VDR, for example, data sent using IEC 61162-450.



Click “Save” and “Apply changes”

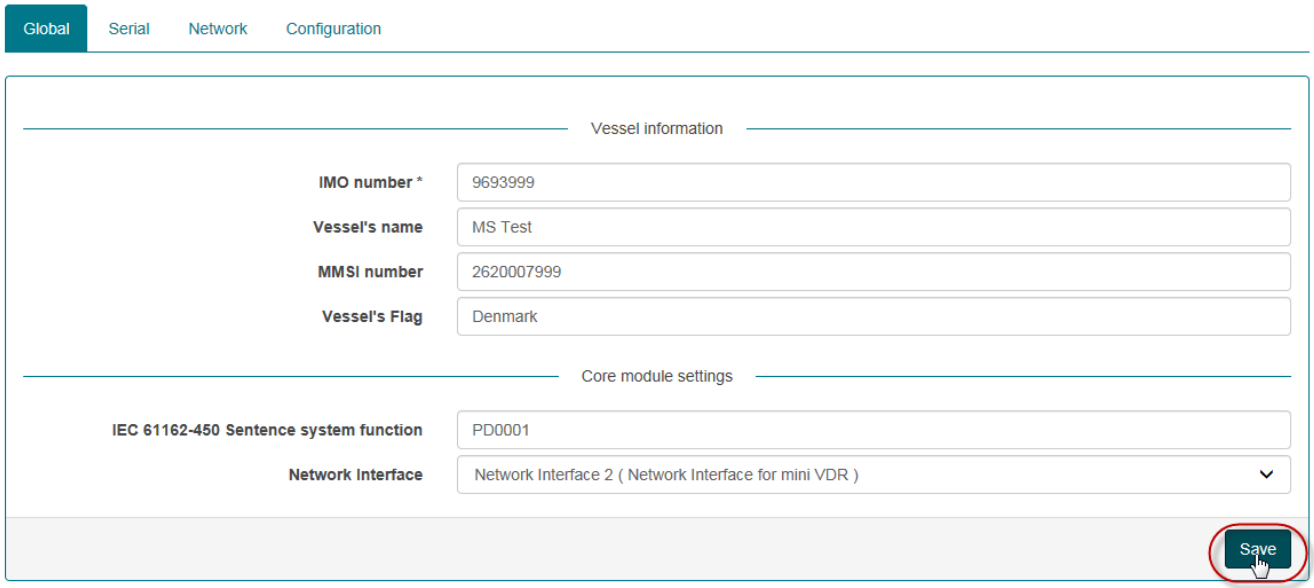
Check that the network configuration is correct.
Click “Apply Changes”.

You may be prompted to set up the global parameters for the mini VDR, see section 6.4.

6.4 Configuration of global parameters for the Mini VDR

Go to “Mini VDR -> Global”.

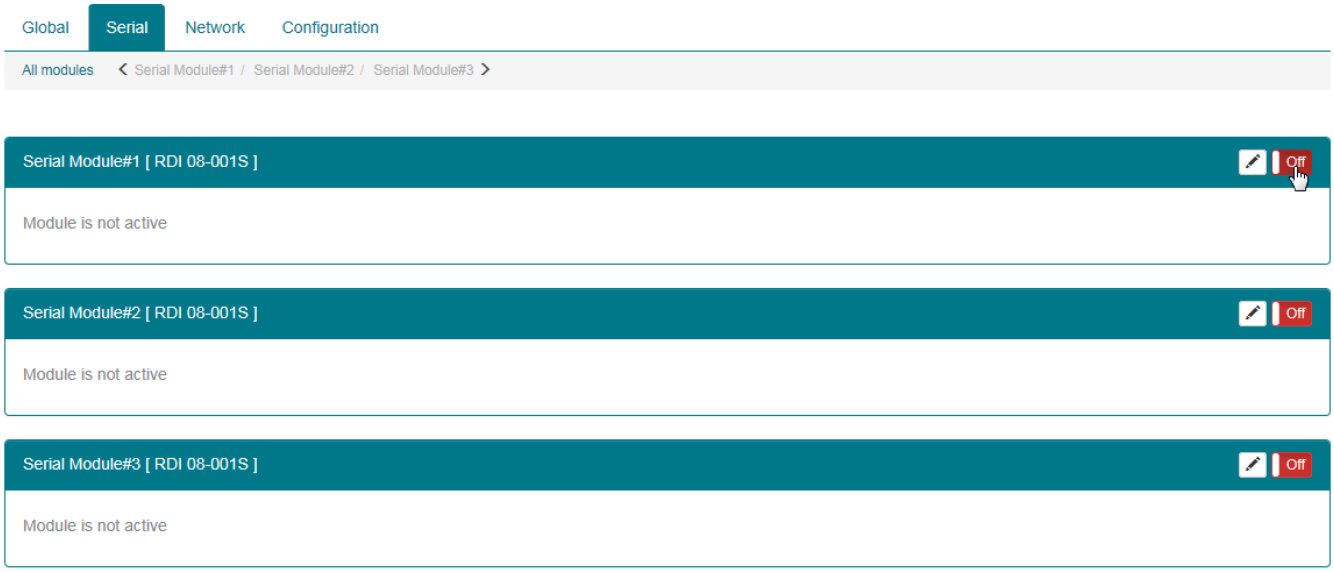
Enter vessel information.
Configure the IEC 61162-450 Sentence system function (default is PD0001)
Assign a network interface for the mini VDR.



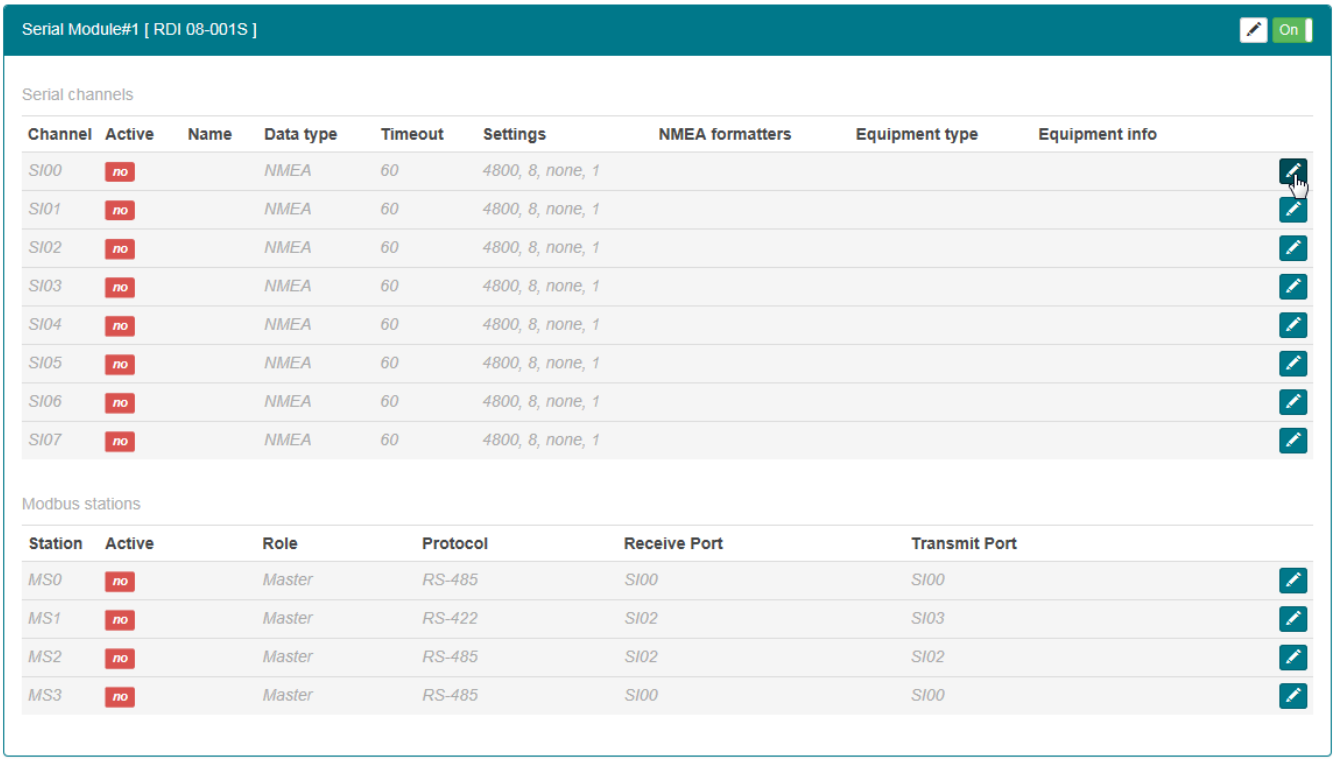
Click “Save” and, if required, reboot the VRS (check the yellow bar at the top of the page).

6.5 Configuration of serial modules

Go to “Mini VDR -> Serial”.



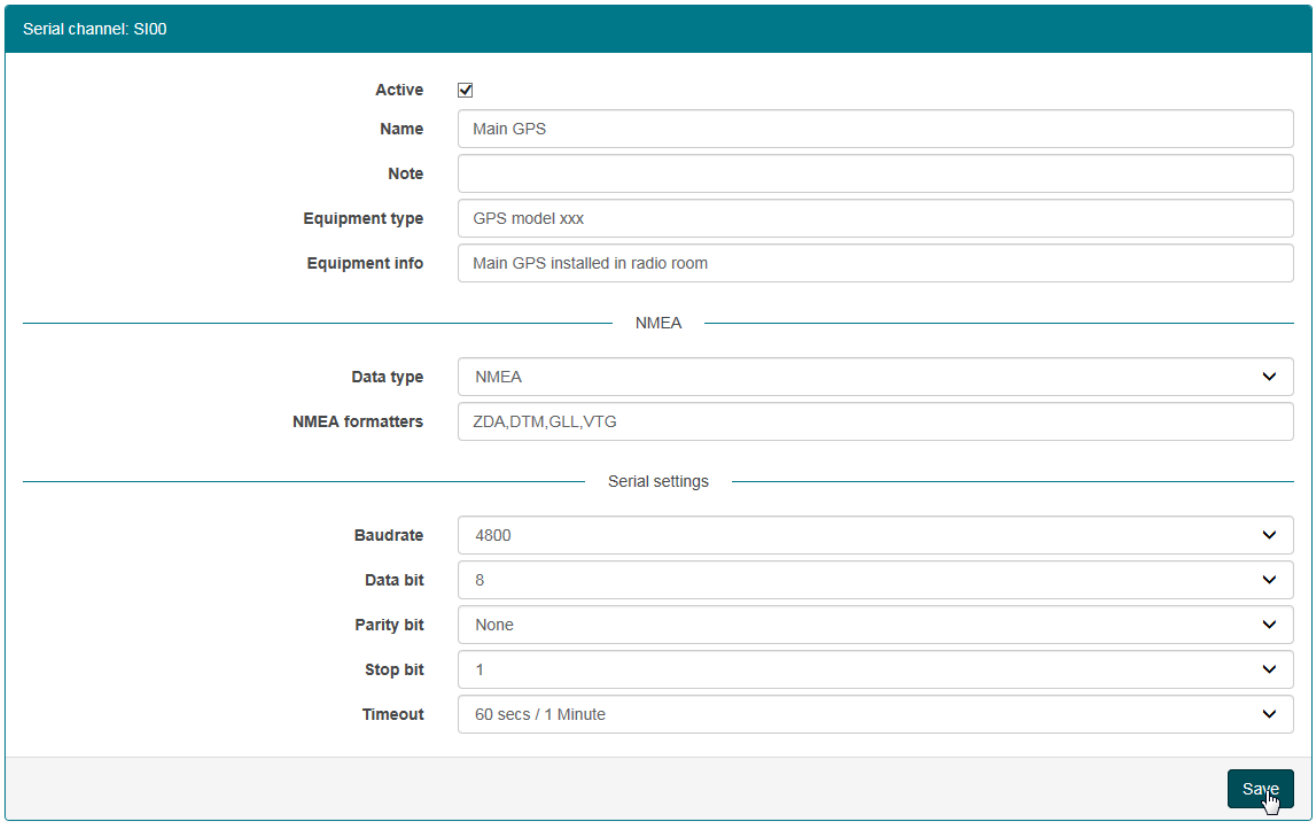
Enable serial module #1



The serial interfaces on the RDI 08-001S may now be configured.

6.5.1 Configuration of a serial interface

The communication parameters and other parameters must be set for interfaces in use on the RDI.



Click on “Save” to apply the configuration.

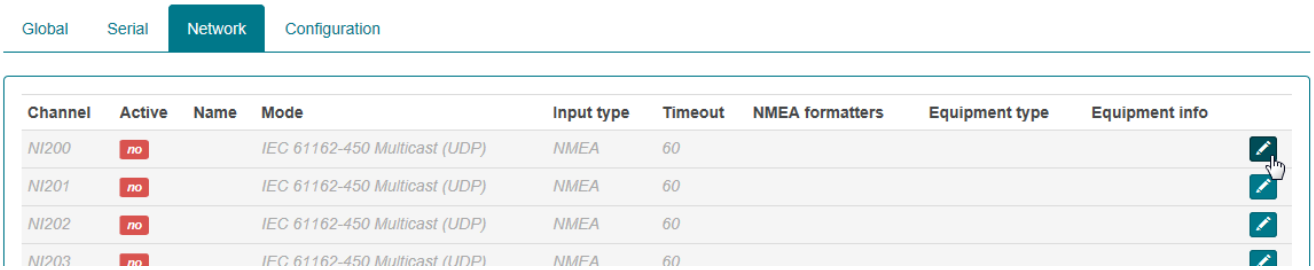
6.5.2 Description of descriptive fields

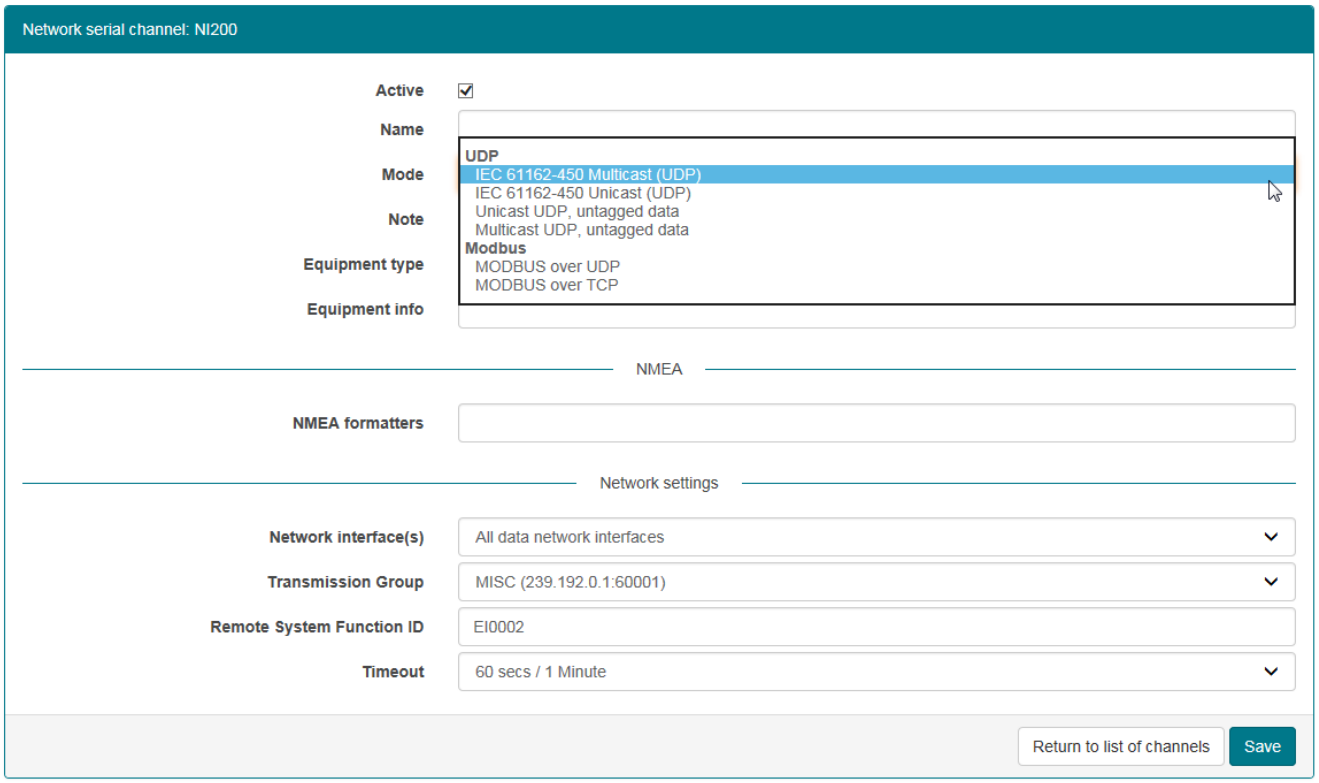
Label in VRS configuration I/F	Label in VDR configuration I/F	Description
Name	Channel ID	This text will be used a “channel label” in the VDR Explorer.
Note	Note	Free text.
Equipment type	Source type	Enter an unambiguous description of the connected equipment here e.g. type, model.
Equipment info	Source	Additional description of the equipment may be entered here.
NMEA formatters	List of NMEA formatters	When relevant enter a list of the NMEA formatters for the sentences to be received on the interface. This may be useful for troubleshooting.

6.6 Configuration of interface for receipt of network data

The mini VDR can receive data from the network. Several protocols, including IEC 61162-450, are supported.

Go to “Mini VDR->Network”.

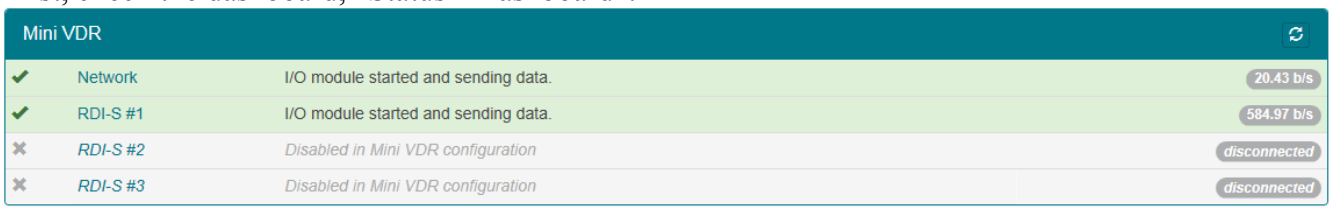




Activate the channel, select mode, set up the required parameters and click on “Save”.

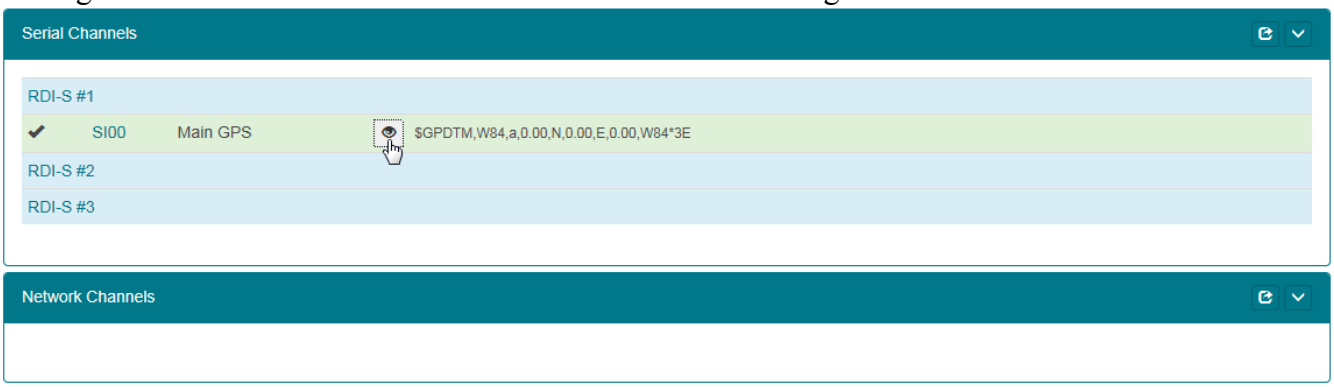
6.7 Status information related to RDIs

First, check the dashboard, “Status->Dashboard”.



The mini VDR section shows whether an RDI is connected. In this case, RDI-S #1 is connected.

Then go to “Status->Mini VDR” and check whether data is being received.



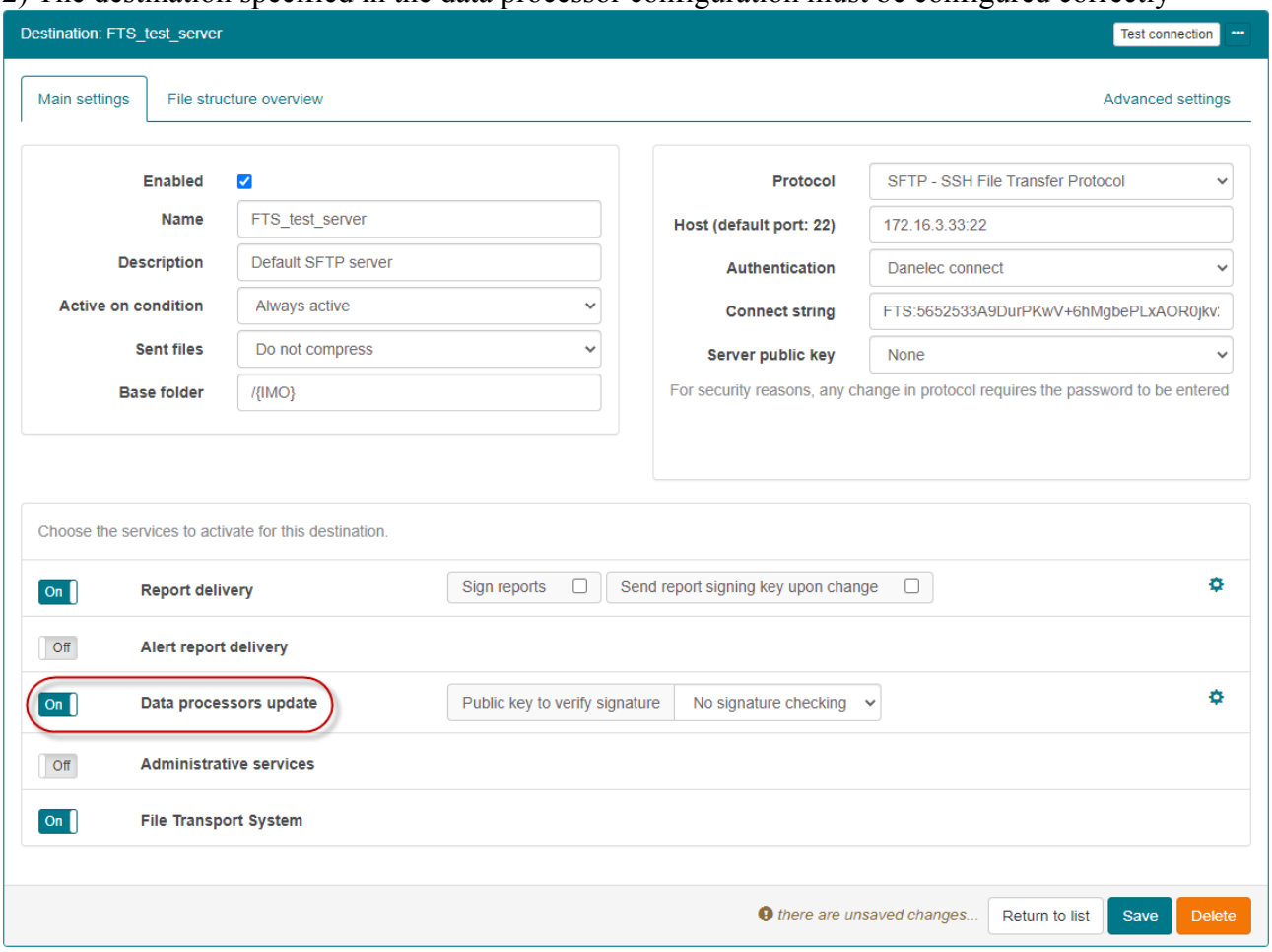
Click on the “eye symbol” to see the details.

7 Remote configuration of data processors

Rather than manually updating the data processor configuration(s) in the VRS, an administrative server may be used for that purpose. This means that the configuration files for each VRS must be uploaded to the administrative server, and the VRS will check the administrative server regularly. This scheme also works for topologies where a VRS is located behind a firewall, which prevents remote access to the VRS from shore.

1) The data processor must be configured accordingly see section 7.1.1

2) The destination specified in the data processor configuration must be configured correctly



The directory structure and the filenames on the server must apply to the principles as described in section 4.4.2. However, only the strings IMO, VESSELNAME, DEVICEID, and DATAPROCESSORID are valid keys.

The VRS may be configured accordingly by selecting “Advanced settings -> Data processor update”. Unless there is a good reason, do not change the default configuration.

An example of this is shown below:

- IMO number: 1239999
- {BASE} is /IMO

- Data processor name: Report_generator_1
- The data processor names Report_generator_1 is configured to request a remote update

The default definition for configuration files is: {BASE}/configs/{DATAPROCESSORID}.vec

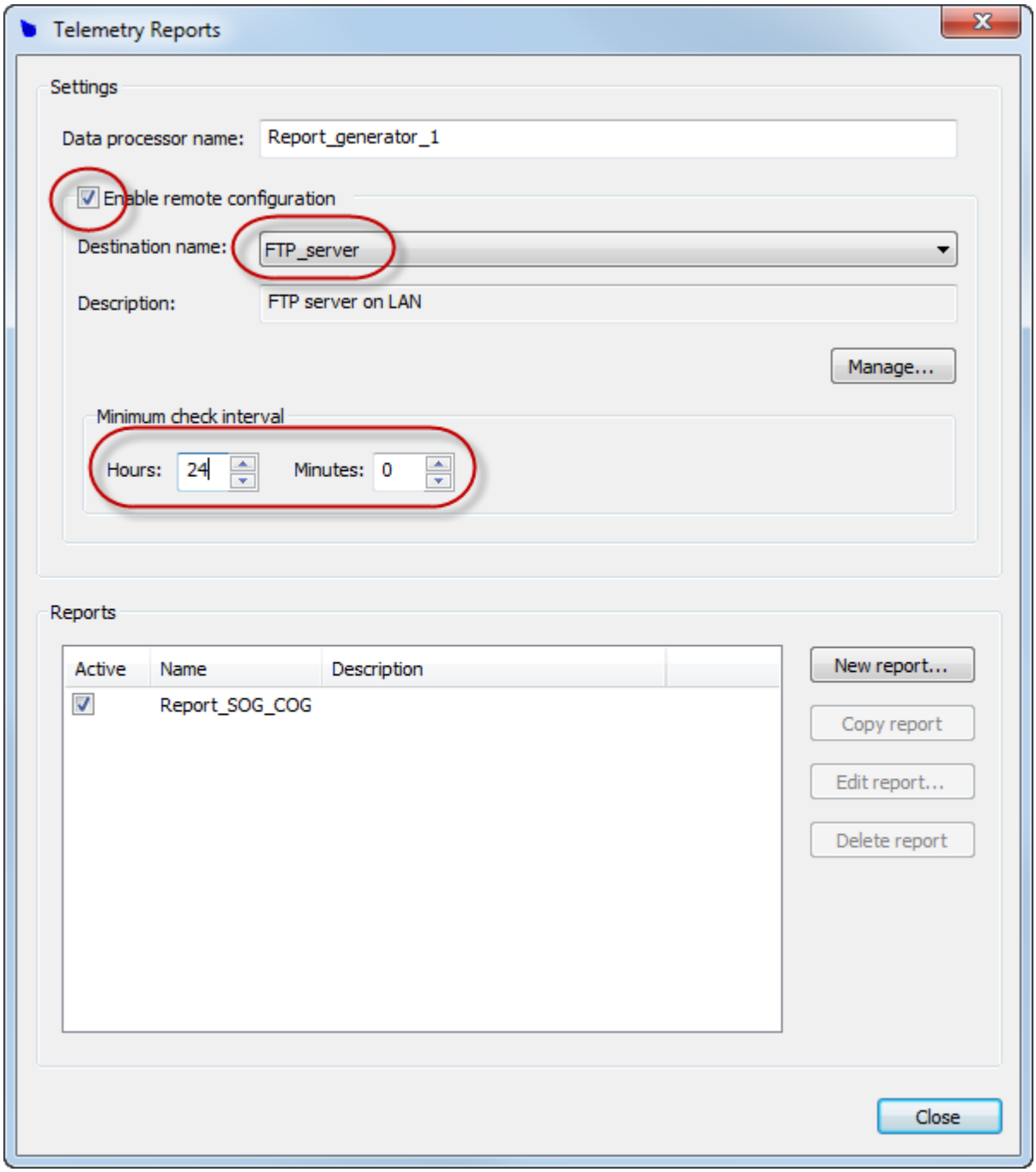
In this case the VRS search for a file named:

/1239999/configs/Report_generator_1.vec

If the file is present on the server, it will be downloaded, applied, and renamed. The VRS will return a status report to the server showing whether the update was successful.

7.1.1 Configuring a data processor for requesting a remote update

As described in section 5, the VDR Explorer is used for creating a data processor configuration.



Check the box “Control connection”, select a destination and configure “Minimum check interval”. Note that the control server may be an independent server that is not used as a destination for reports.

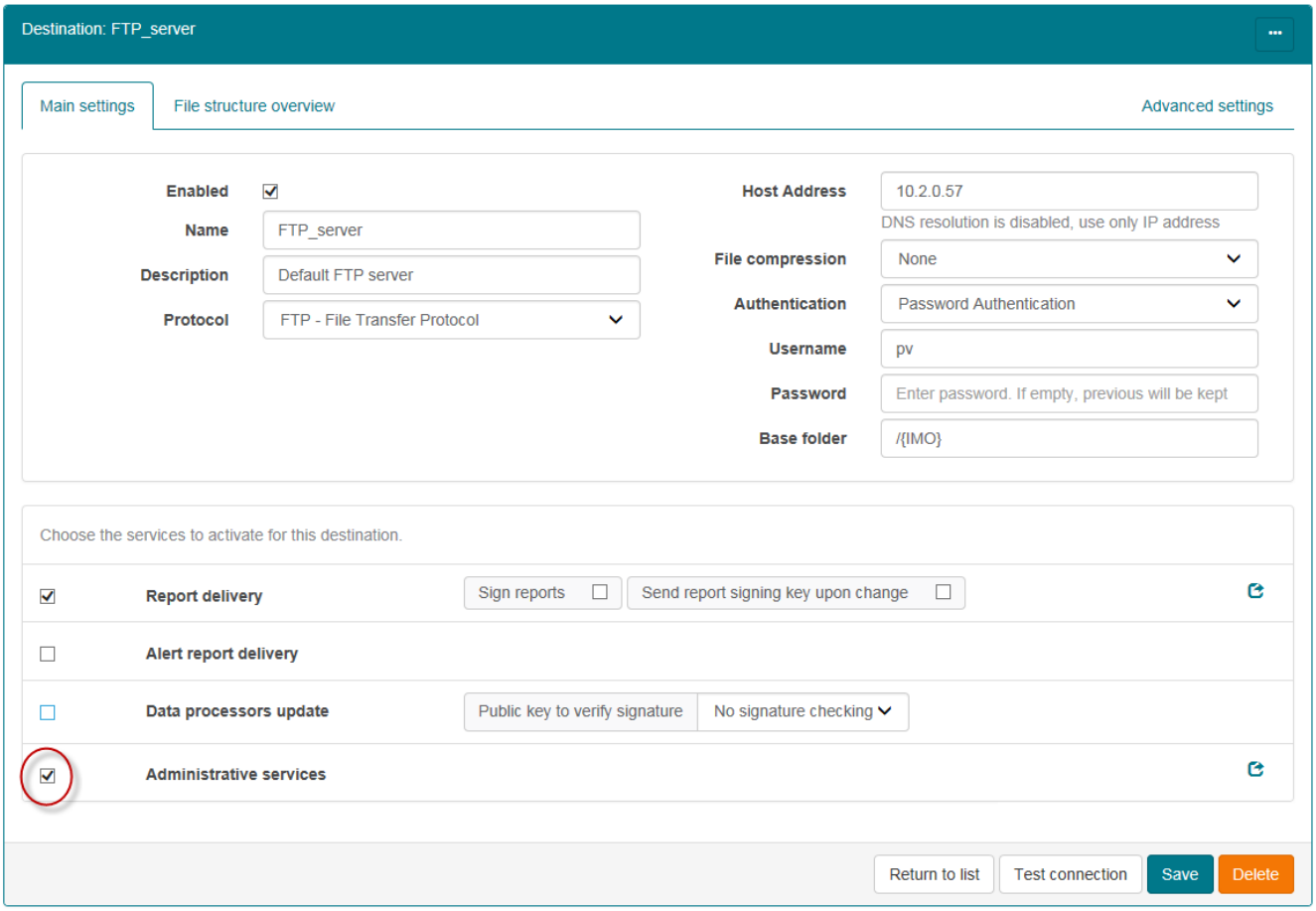
8 Remote administration of the VRS

A VRS may be configured to download and apply the following items from an “administrative server”.

- Firmware
- License file
- VRS configuration
- Remote commands

The basic principle is that the VRS will contact the administrative server regularly or when the VRS front panel switch is changed into “EXTENDED ACCESS” position, and search for a specific file. If present, the file will be downloaded and renamed. In addition, the VRS will respond with a file containing status information, for example, whether the file passed verification. The file will be deployed by the VRS if it passed verification.

8.1 Preparing a destination for the administrative server

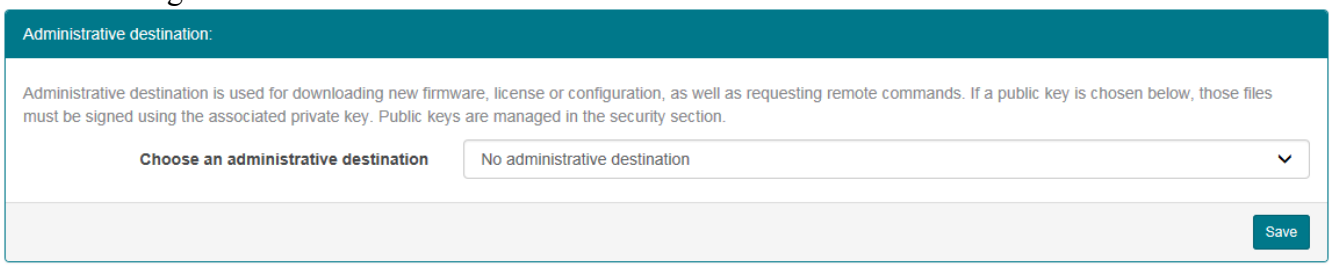


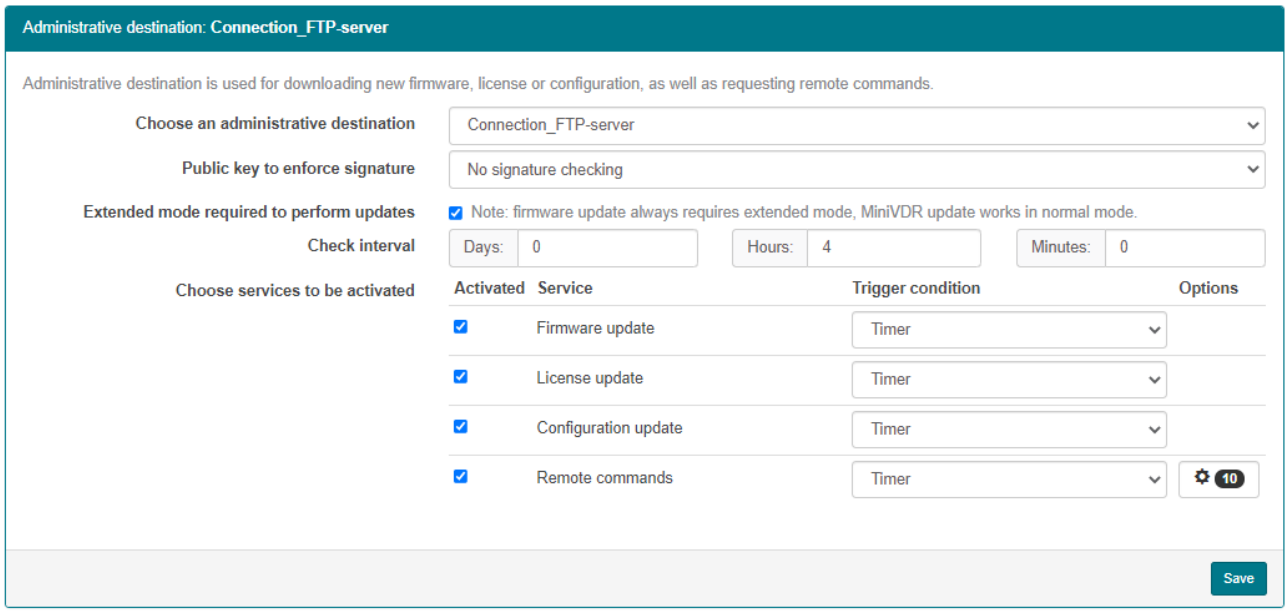
The destination to be used for the “administrative server” must be configured accordingly.

8.2 Selecting the destination for the administrative server

In addition, the destination for the “administrative server” must be designated.

Go to "Configuration->Destinations"





Select destination, configure relevant parameters,” and click on “save”.

8.3 Remote firmware update

The VRS will when triggered, attempt to download the following file from the server:

{BASE}/firmware/vrs2_firmware.bdf

Per default {BASE} is /IMO but it is configurable, see section 4.4.1 "Base folder".

8.4 Remote license file update

The VRS will when triggered, attempt to download the following file from the server

{BASE}/license/softwarelicense.bdf

Per default {BASE} is /IMO but it is configurable, see section 4.4.1 "Base folder".

8.5 Remote configuration update

The VRS configuration is divided into three parts, which may be updated independently

- VRS device configuration
- Configurations file for destinations
- Mini VDR configuration

The VRS will, when triggered, attempt to download the following files from the server

{BASE}/device_config/configuration.xml
{BASE}/destination_config/destinations.xml
{BASE}/minivdr_config/minivdr.xml

Per default {BASE} is /IMO but it is configurable, see section 4.4.1 "Base folder".

8.6 Remote commands

Remote commands are used for management and debugging purposes.

The VRS will, when triggered, attempt to download the following file from the server

{BASE}/commands/command.xml

Per default {BASE} is /IMO but it is configurable, see section 4.4.1 "Base folder".

10 Additional modules (functionality)

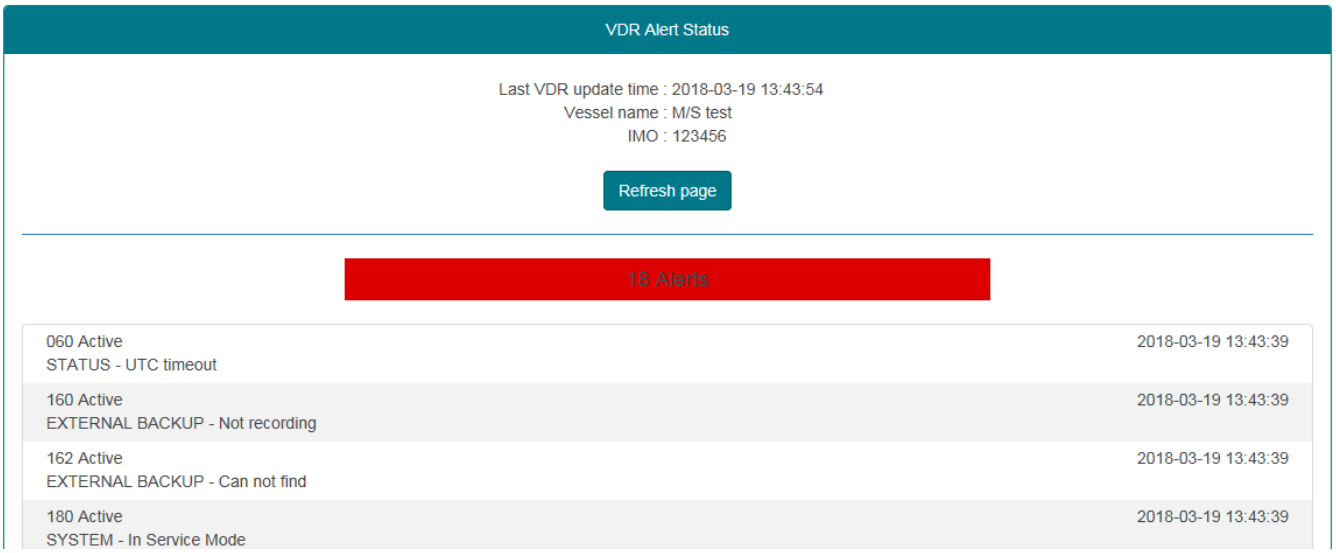
Go to “System->Modules”.

10.1 VDR remote management

See section 2.1.

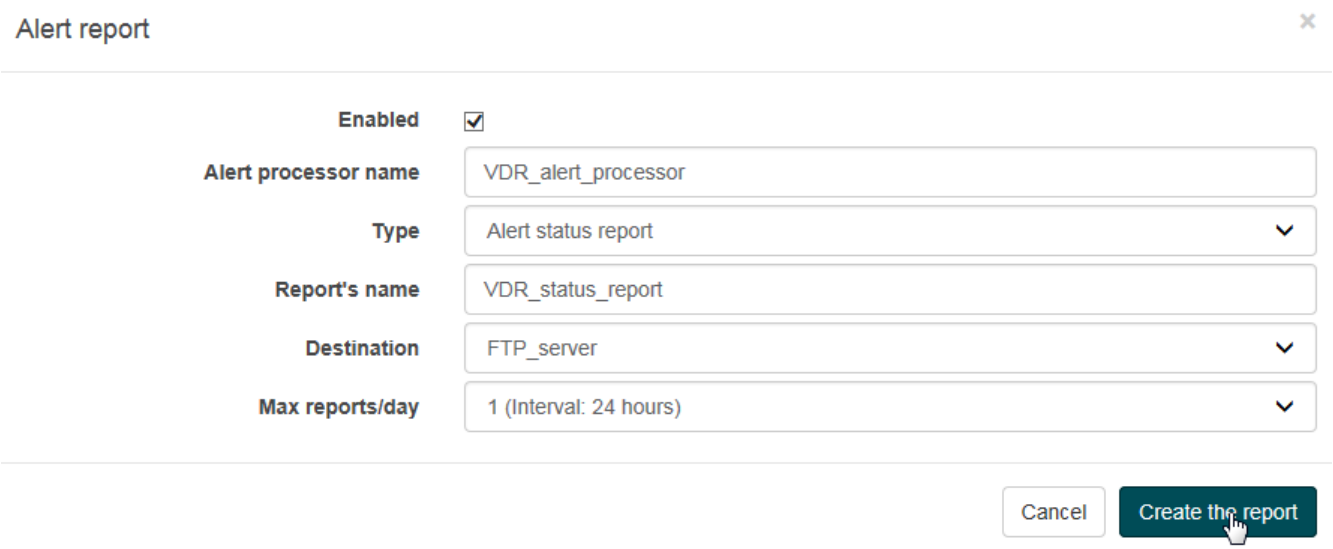
10.2 VDR alert monitoring

If enabled, it is possible to see the current alert status for the VDR.
Go to “Home->VDR Alert Status”.



10.3 VDR Alert Processing

The VRS may be configured to transmit the VDR status as a report to a destination (server).
Go to “Configuration->Alert processing->Add a new report”.



Enter relevant parameters and click on “Create the report”.

10.4 Data collection

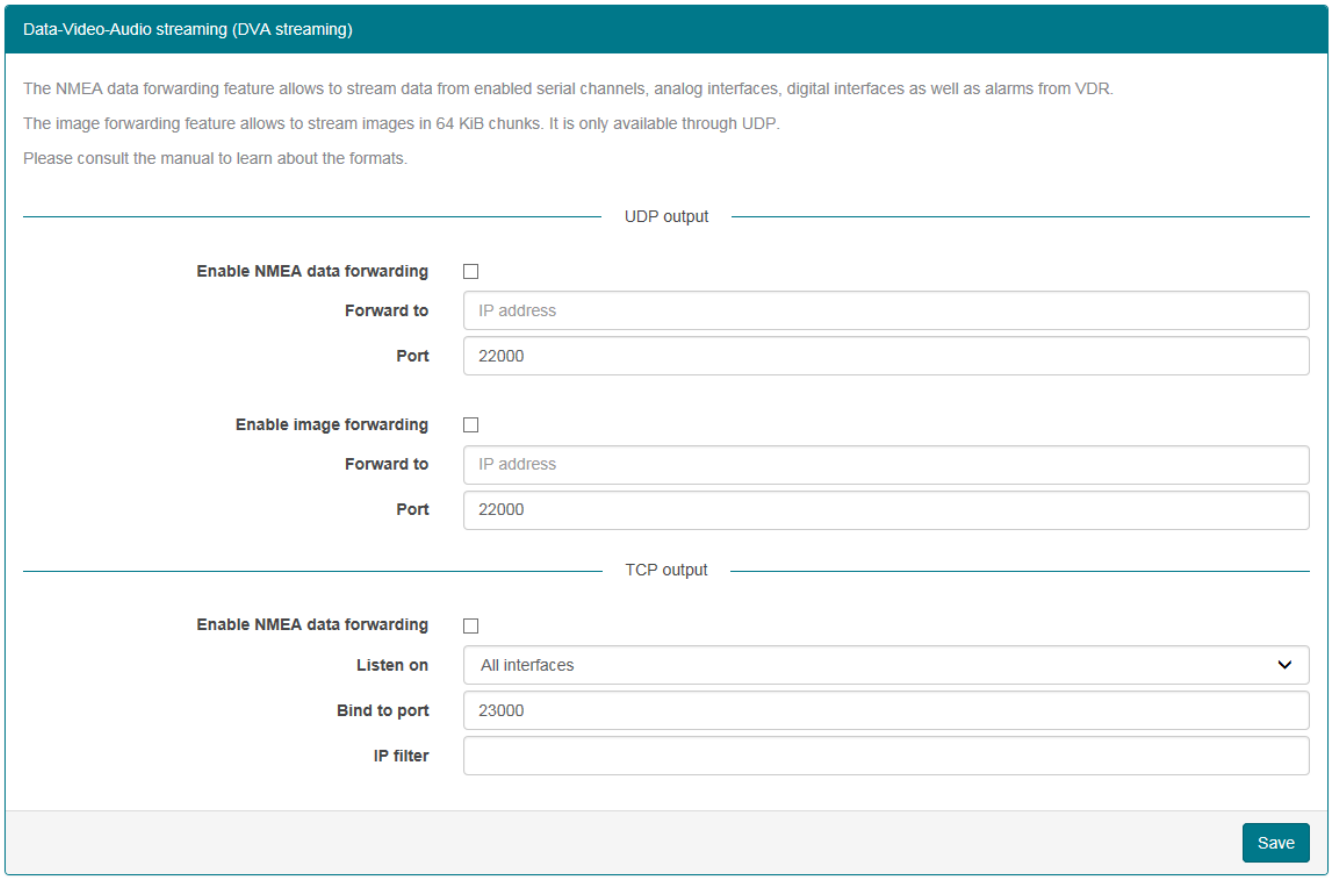
Data processor functionality may be enabled or disabled entirely; see section 5.

10.5 DVA streaming

The VRS can stream VDR data (live - as they are recorded) to other systems.
Currently, the following data may be streamed:

- Serial data
- Analog data decoded as IEC 61162-1 proprietary strings
- Digital data decoded as IEC 61162-1 proprietary strings
- Image data
- Alert information from the VDR decoded as IEC 61162-1 proprietary strings

DVA streaming is configured by going to “Configuration->DVA streaming”.


Data-Video-Audio streaming (DVA streaming)
The NMEA data forwarding feature allows to stream data from enabled serial channels, analog interfaces, digital interfaces as well as alarms from VDR.
The image forwarding feature allows to stream images in 64 KiB chunks. It is only available through UDP.
Please consult the manual to learn about the formats.
UDP output
Enable NMEA data forwarding
Forward to
IP address
Port
22000
Enable image forwarding
Forward to
IP address
Port
22000
TCP output
Enable NMEA data forwarding
Listen on
All interfaces
Bind to port
23000
IP filter
Save

10.6 System monitoring

The VRS may be configured to send system information to an MQTT broker; see section 15.

10.7 Hosting of applications

The VRS may host APPs, which can handle data formats and carry out calculations that a data processor cannot do, see section 16.

10.8 File Transport System, FTS

A VRS may be used for transporting files to and from shore in a controlled, and secure manner see section 14. The functionality is only for applicable special OEM solutions that are utilizing the VRS and Danelec cloud services.

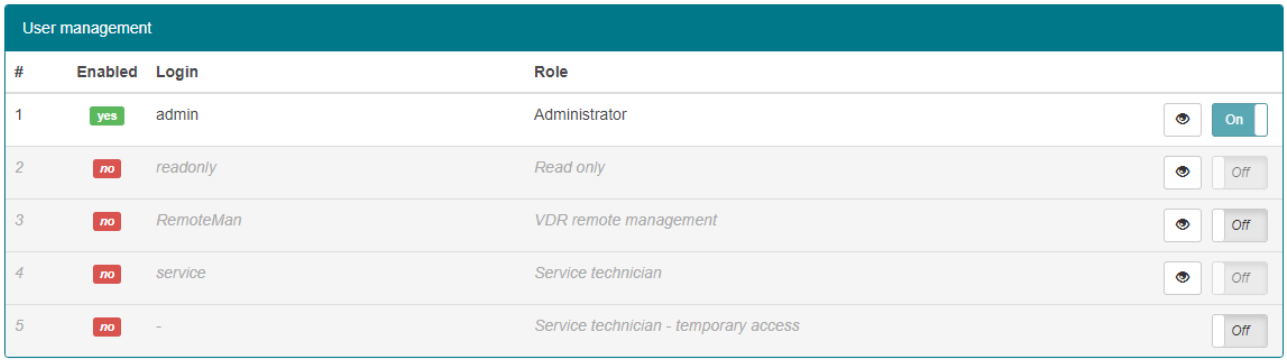
10.9 TeamViewer support

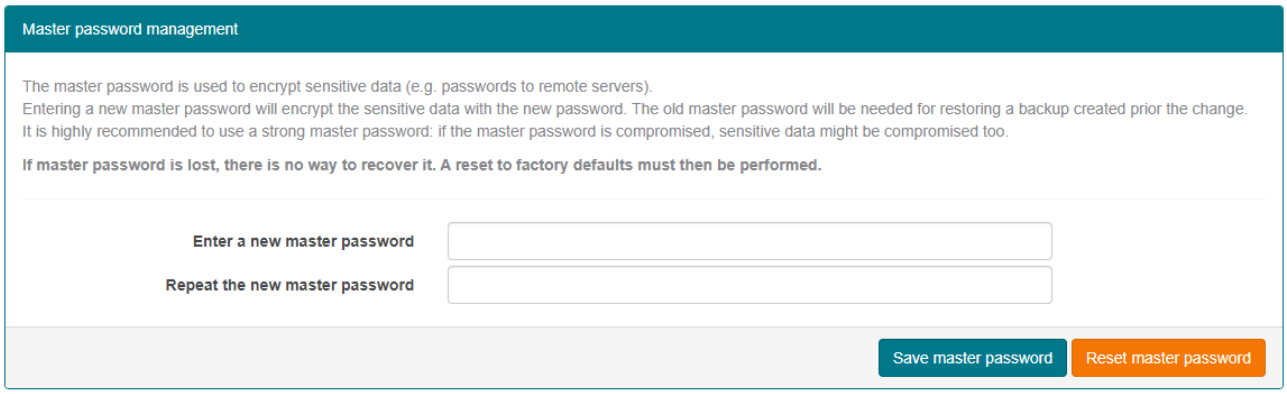
A TeamViewer connection may be used for remote management of Danelec VDRs as an alternative to, for example, VPN connections. For more information, see MAN13735 VRS remote management TeamViewer manual.

11 Maintenance and administration

11.1 User and password administration

Go to “System->Users”.





11.1.1 User administration

11.1.1.1 Administrator

The administrator (admin) cannot be disabled. The default password for the administrator must be changed as part of the initial installation.

11.1.1.2 Read-only

The read-only user will not be able to change any parameters. Note that the read-only user cannot be enabled until a password is set.

11.1.1.3 VDR Remote Management

The remote management user is required for remote VDR management. Note that the remote management cannot be enabled until a password is set.

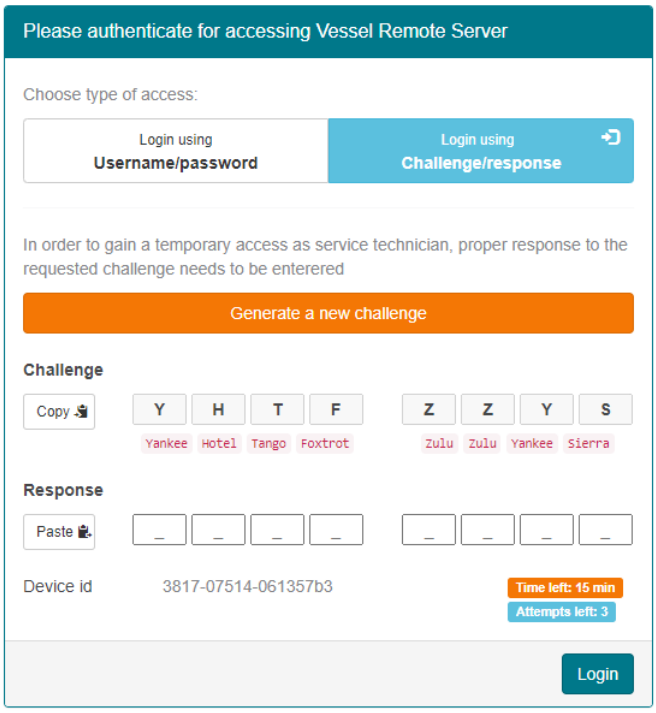
11.1.1.4 Service technician

The service user may read and change all parameters except for:
Parameters related to user administration
The master password

Note that the service user cannot be enabled until a password is set.

11.1.1.5 Service technician - temporally access

Exposing a “permanent” password is a security risk. An alternative to this is exposing a temporary password. For the VRS, a challenge-response algorithm has been implemented. When enabled, the login page will change.



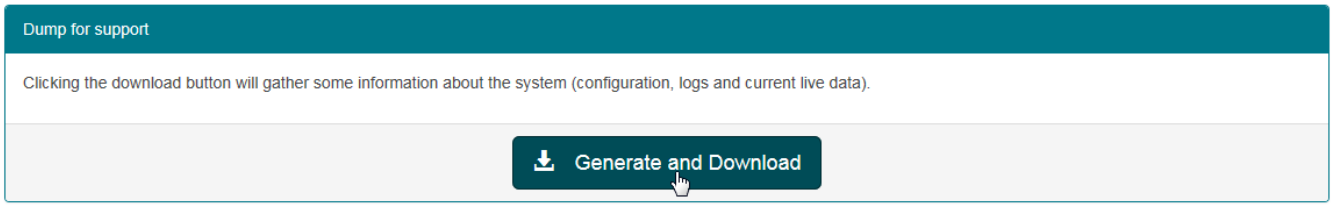
The challenge must be sent to the system administrator, who will reply with a response. Creating the response requires an application for a Windows PC and the master password; see MAN13426 “User Manual for the VRS password and security tool”.

11.1.2 Master password

The master password is required to encrypt sensitive data and is also required for the challenge-response algorithm.

11.2 Dump for support

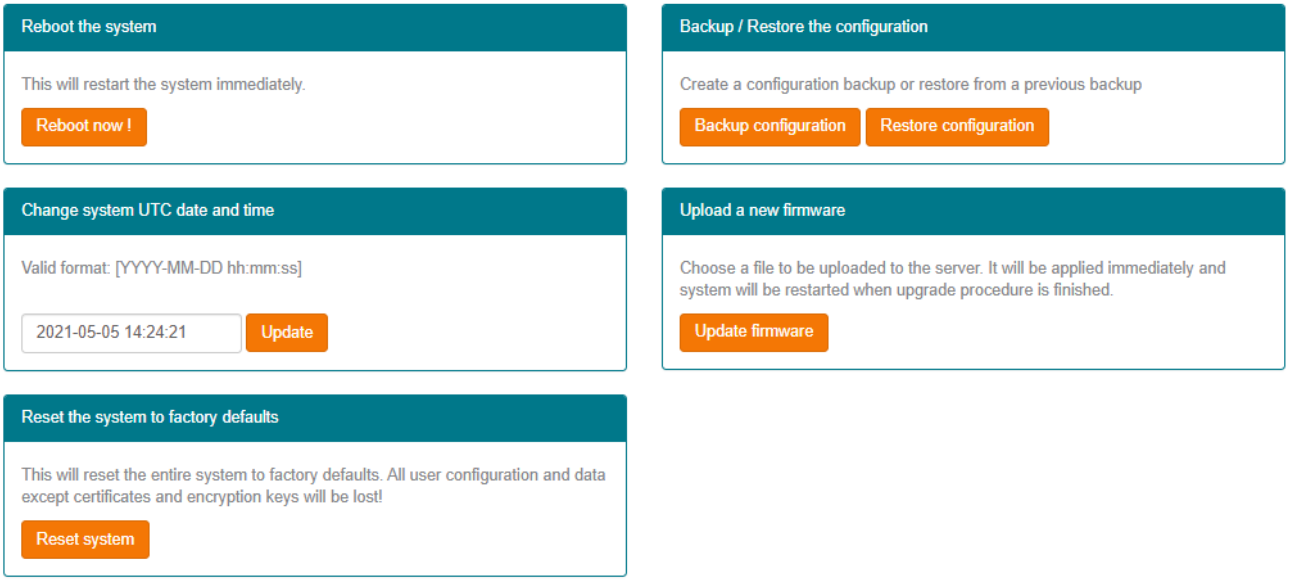
Go to “System->Dump”.



A “Dump for Support” may be required for troubleshooting by Danelec Marine Support.

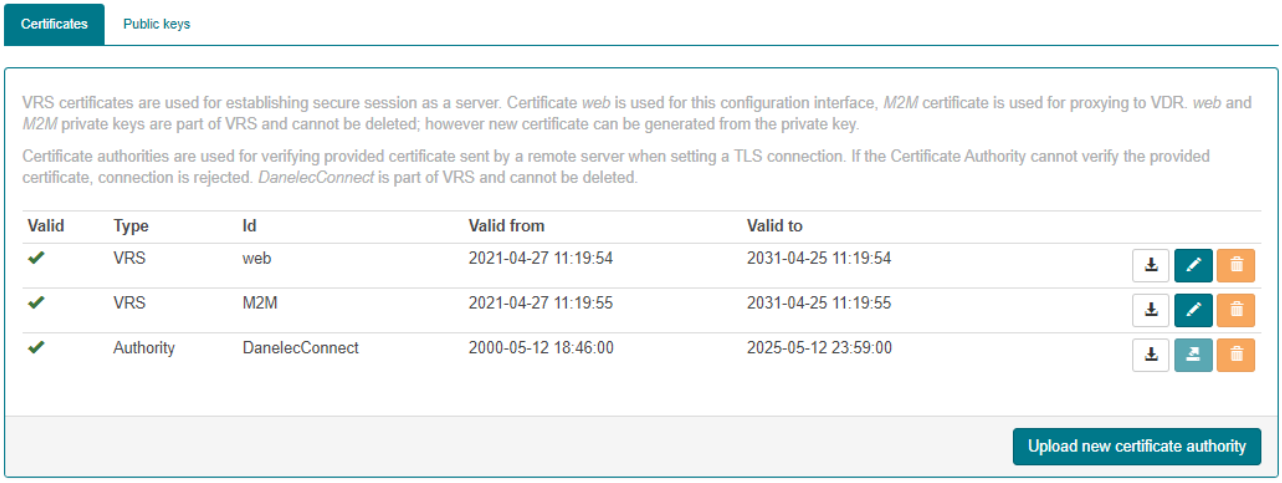
11.3 System maintenance

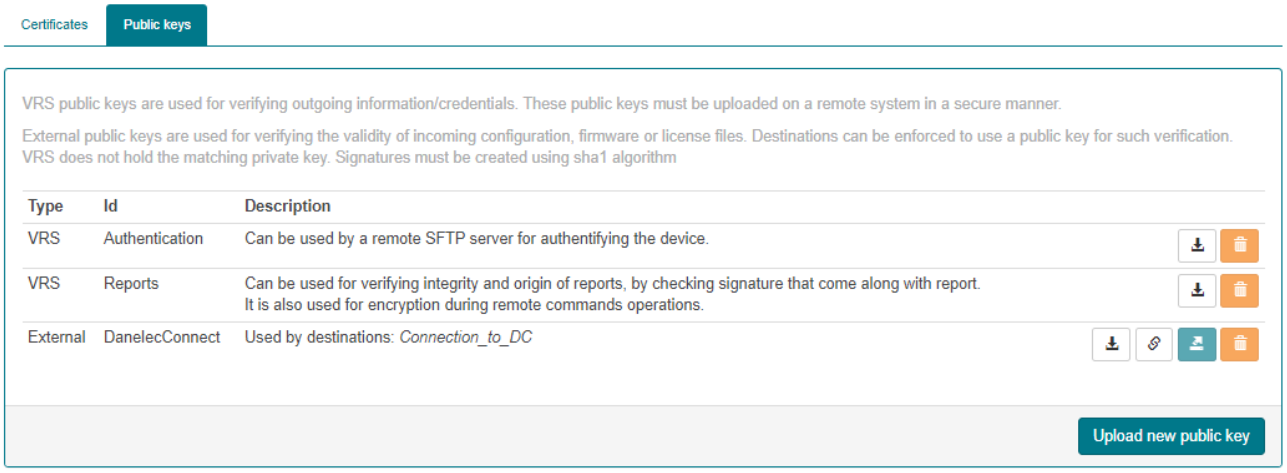
Go to “System->Maintenance”.



12 Additional security, public keys

The VRS uses different kinds of cryptography and certificates to increase data security.
Go to “System->Security”.





12.1 Certificates

These certificates are required to establish encrypted connections. In addition, see section 12.4.

12.2 VRS public keys

These public keys may be used as authentication for a VRS when it logins to a destination server and for verifying signed files sent by the VRS, see section 4.3. These public keys must be downloaded and securely transferred to the server and deployed there.

12.3 External public keys

Configuration files sent to the VRS may be signed. The public key(s) required for verifying the files must be uploaded to the VRS.

One public key may be deployed for verifying data processors configuration updates; this may be enabled when the destination for the data processor is being configured see section 4.4
The same or another public key may be deployed for verifying data related to remote administration of the VRS see section 8.2.

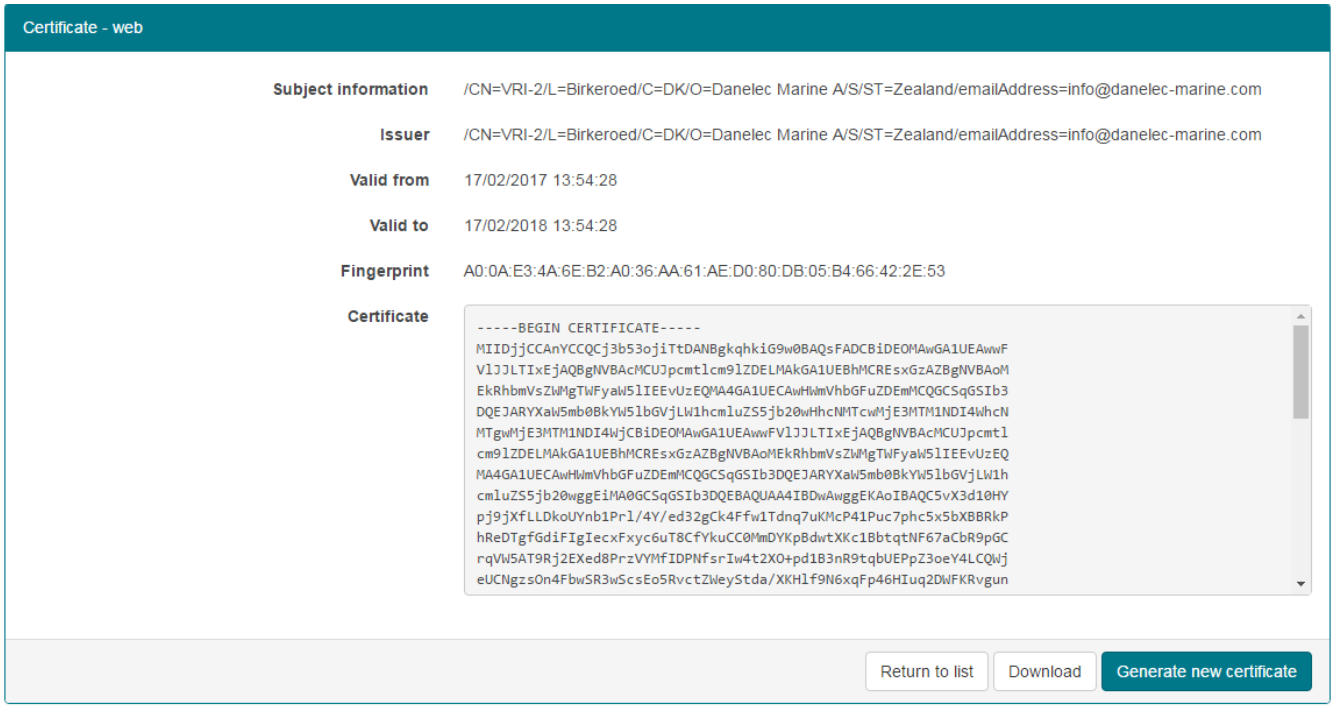
12.4 Management of configuration web and M2M certificates

By default, the VRS creates self-signed certificates for the configuration website and the machine-to-machine interface. These certificates are used to authenticate the configuration website and M2M interface towards the user.

It is possible to change these certificates either by updating the certificate subject information and creating a new self-signed certificate with updated subject information or by generating a Certificate Sign Request (CSR) with updated subject information for external 3rd party signing.

Select the menu point “System->Security” to get a list of certificates that can be downloaded and edited. Click on the download icon to download the certificate in .pem format. The downloaded certificate can then be imported into, for example, a certificate store of “trusted sites” of a web browser or 3rd party tool using the M2M interface.

Click the edit icon of the certificate to view and edit the certificate details:

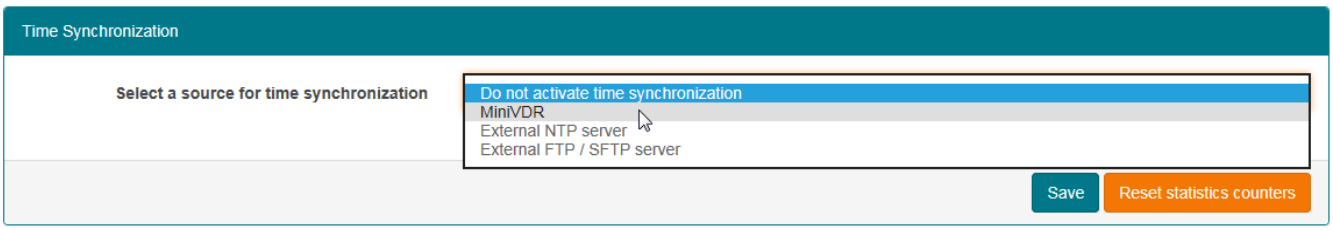


Click the “Generate new certificate” button to start editing the certificate subject information and duration:

13 Time synchronization

For most applications, it is important that the timestamp for data reports and other information is accurate. The VRS has a built-in “clock”, but unless synchronized from an outside source, this “internal time” will become inaccurate.

Go to "Configuration -> Time Synchronization"



Select a source.

In general, the VDR or MiniVDR must be used as a time synchronization source, assuming that a GPS is connected to them.

External FTP/SFTP server is the least desirable source since data will have to be transmitted regularly, and it is less accurate.

The following file will be created on the server if External FTP/SFTP is used as a synchronization source.

{BASE}/utcsynch

Per default {BASE} is /IMO but it is configurable, see section 4.4.1 "Base folder".

14 FTS file transfer

The VRS facilitates the transfer of files to and from shore for third-party applications and equipment. The interface for the client on the VRS is a JSON/REST interface which is described in DBS12436 “VRS 002 File Transport System REST/JSON API”. FTS is a service that may be used for (for example) the exchange of routes and noon reports.

Danelec Marine has developed an application (FTS test tool) for a Windows PC, which may be used for testing the interface.

Before FTS may be used, a license is required, and the FTS module must be enabled (System->Modules).

14.1 Configuration of FTS

The following steps must be carried out:
1) Upload of license file for FTS
2) Enabling the FTS module
3) Configuration of destination for FTS
4) Configuration of FTS
5) Test of FTS (Optional)

14.1.1 Upload of a license file

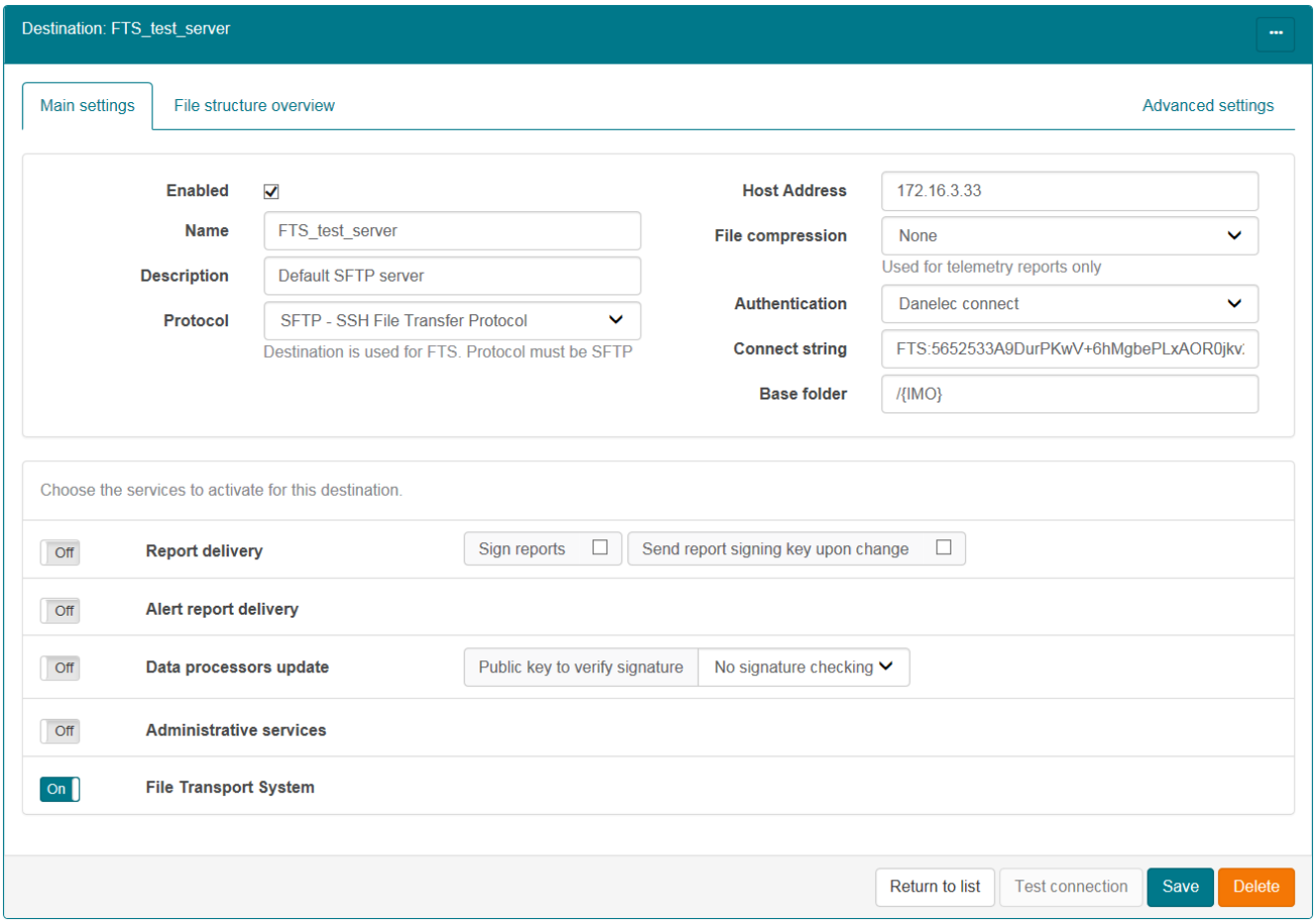
Before FTS may be used, a license is required; go to System->Licenses.

14.1.2 Enabling the FTS module

Before FTS may be used, the FTS module must be enabled; go to System->Modules.

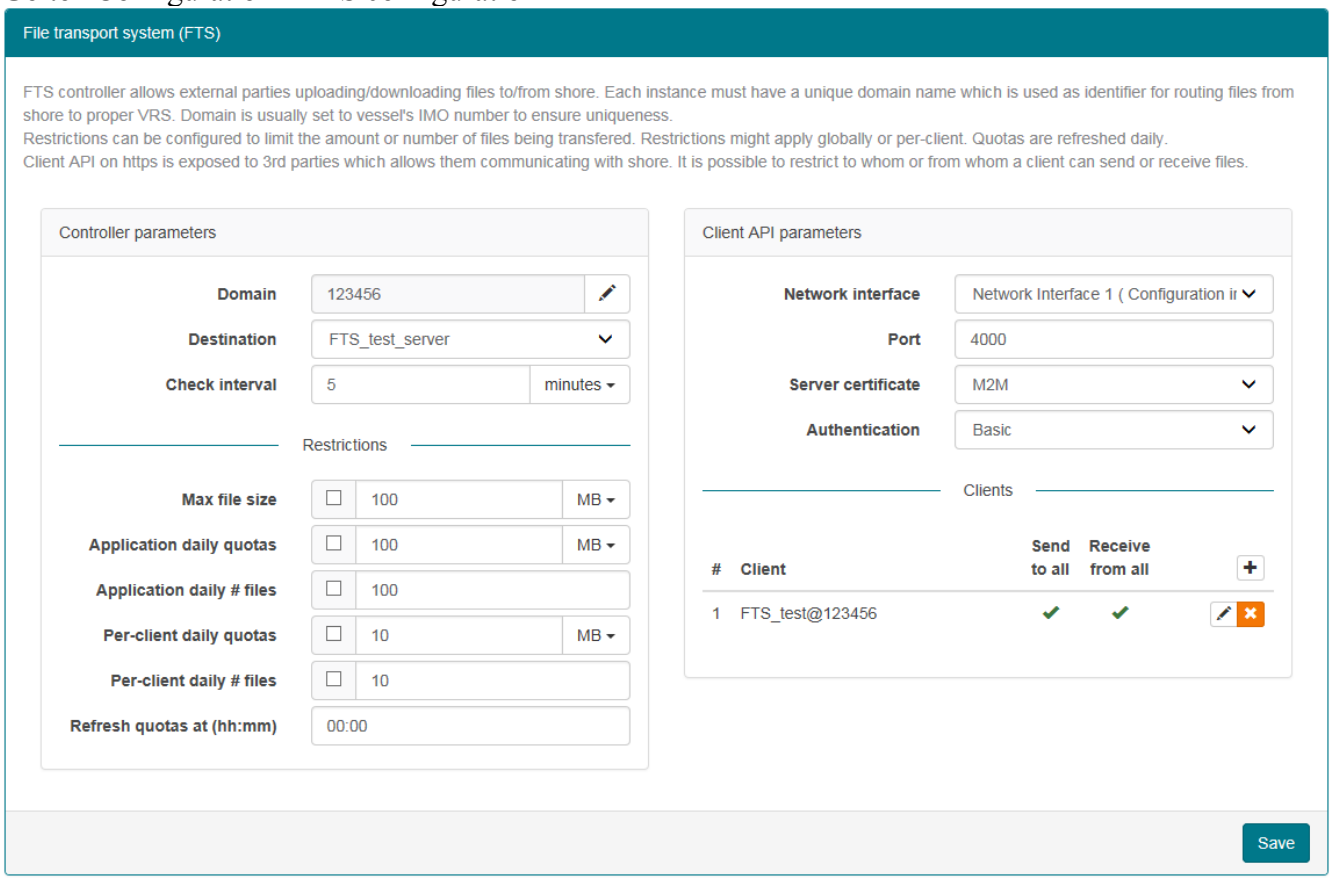
14.2 Configuration of destination for FTS

FTS requires that a destination to a server is configured; go to Configuration->Destinations.



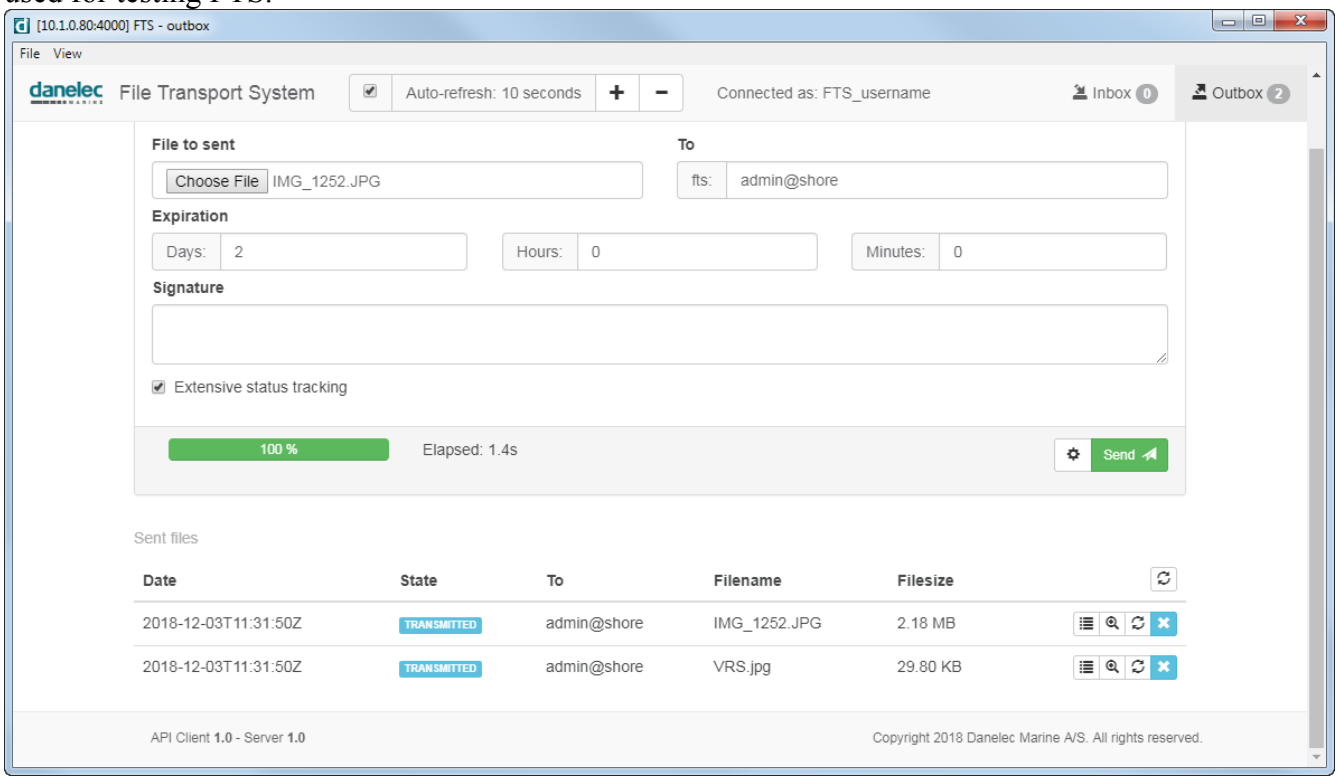
14.2.1 Configuration of FTS

Go to "Configuration->FTS configuration"



14.2.2 Test of FTS

Danelec Marine has developed the “FTS test tool” application for a Windows PC, which may be used for testing FTS.



14.2.3 FTS status

Go to Status->FTS

Destination	FTS_test_server
Last file received	-
Last file sent	-
Last status received	-
Last status sent	-
Files ready to be sent	0
Total space for FTS	7.27 GB
Available space for FTS	6.87 GB (94 %)

Usage and quotas

Data transfer and file counters are automatically reset daily at 00:00 (UTC) and upon reconfiguration.

	Received files	Sent files	Daily Transfer	Daily transfer quota	Daily file count	Daily file count quota
Application	0	1	2.18MB	-	1	-
Client: FTS_test	0	1	2.18MB	-	1	-

File state counters

Timestamped entries are added daily or by clicking the refresh button.
Each entry shows a snapshot of the total number of files in the inbox and outbox.
The outbox file counters shows the number of files in each state.

Timestamp	Inbox	Outbox									
(UTC)	Total	Total	Accepted	In progress	Transmitted	Downloaded	Acknowledged	On hold	Expired	Rejected	Failed
2018-12-10T16:04:57Z	0	1	1	0	0	0	0	0	0	0	0
2018-12-10T16:01:07Z	0	0	0	0	0	0	0	0	0	0	0
2018-12-10T15:57:37Z	0	0	0	0	0	0	0	0	0	0	0

ACCEPTED	File has been accepted by the local FTS and will be pushed to remote FTS
IN_PROGRESS	Remote FTS has read metadata, accepted the file and will start to download it
ON_HOLD	Remote FTS has read metadata but will not download the file now; this can be due to quotas being exceeded for current period
REJECTED	Remote FTS has read metadata but will not download the file; this could be due to quotas, the sender not belonging to the receiver's white list or an unknown recipient.
TRANSMITTED	File has been fully received by the remote FTS.
DOWNLOADED	Client has downloaded the file on the remote FTS
EXPIRED	File has expired on the remote FTS and has been deleted by the garbage collector
ACKNOWLEDGED	Client has processed the file and does not need it anymore
FAILURE	System failure, for instance, no connection to FTP server

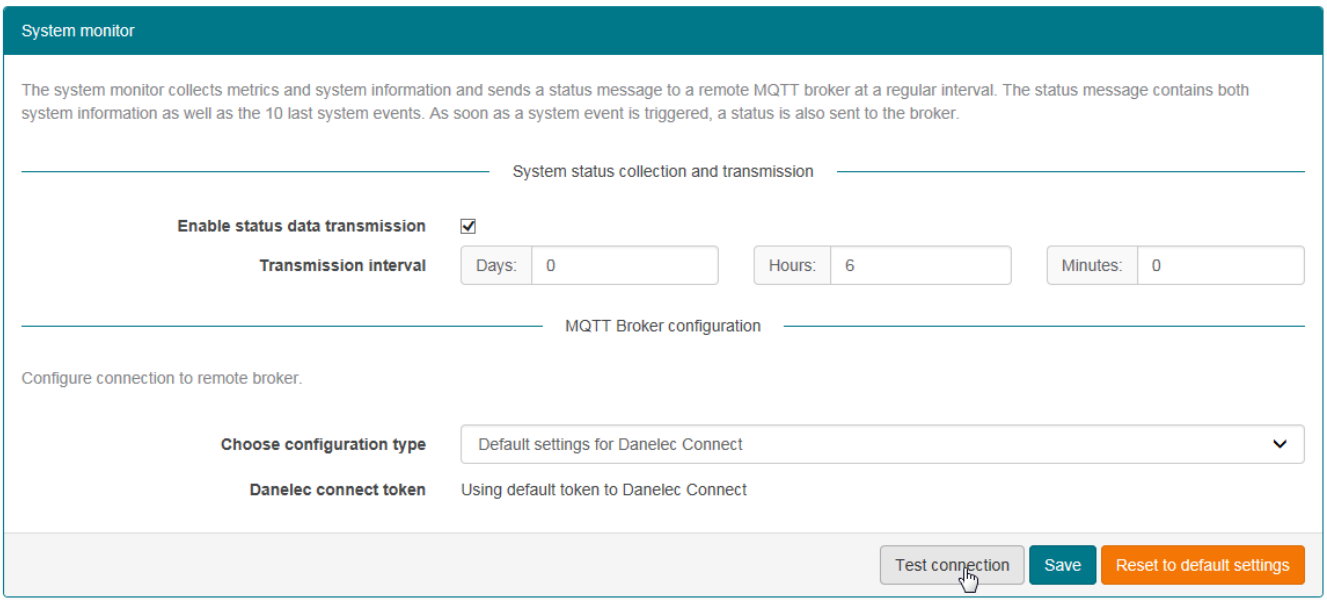
15 System monitor

The system monitor module may be configured to transmit VRS status information to a server. The primary application for this is “semi out of band” system management.

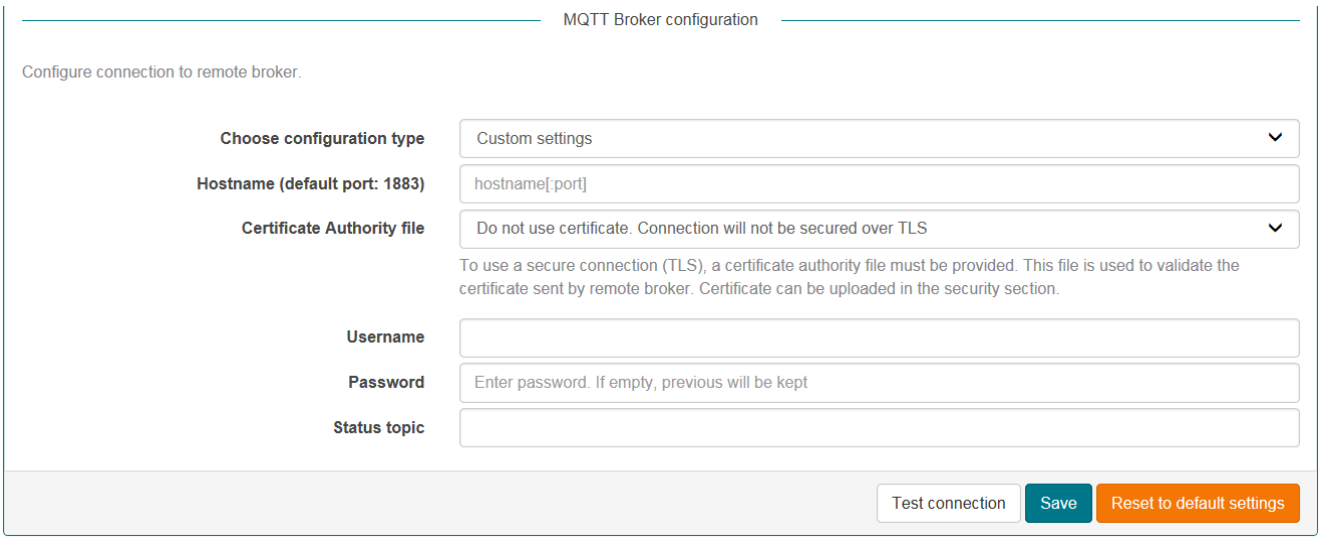
The system monitor is a module that must be enabled; go to System->Modules.

Some parameters for the system monitor are configurable; go to Configuration-> System monitor.

The System monitor uses MQTT as protocol and does not use a “normal” destination (see section 4.4). Per default, the status messages are sent to the Danelec connect server; but another destination may be configured if “Chose configuration type” is set to “Custom Settings”.



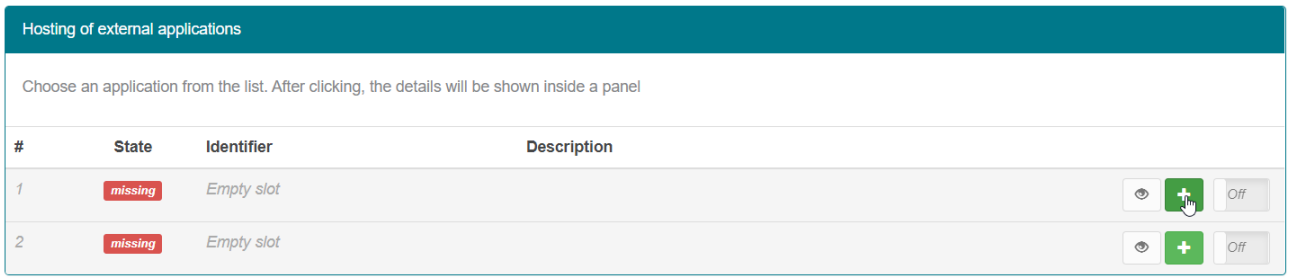
Standard settings for configuration type



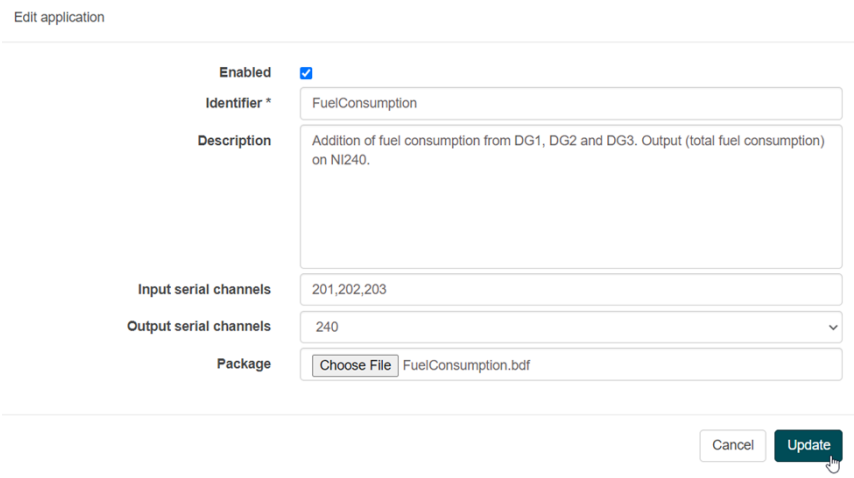
Custom settings for configuration type

16 Hosting of APPs

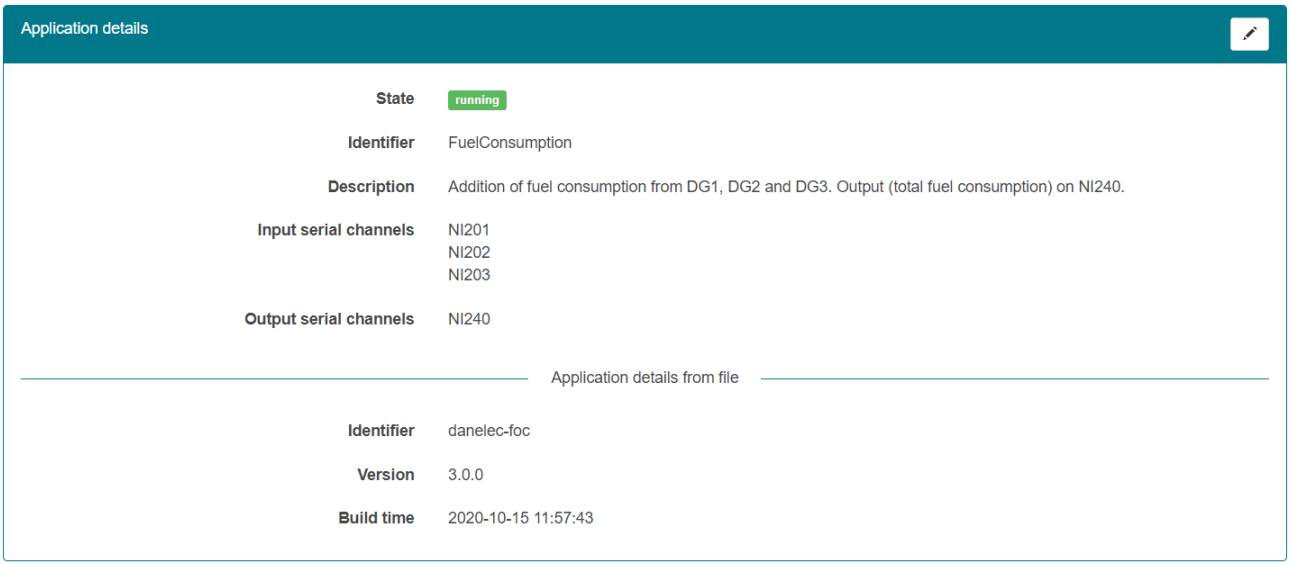
The VRS may host APPs, which can handle data formats and carry out calculations that a data processor cannot do.
Go to Configuration->Application hosting



Select a free application slot

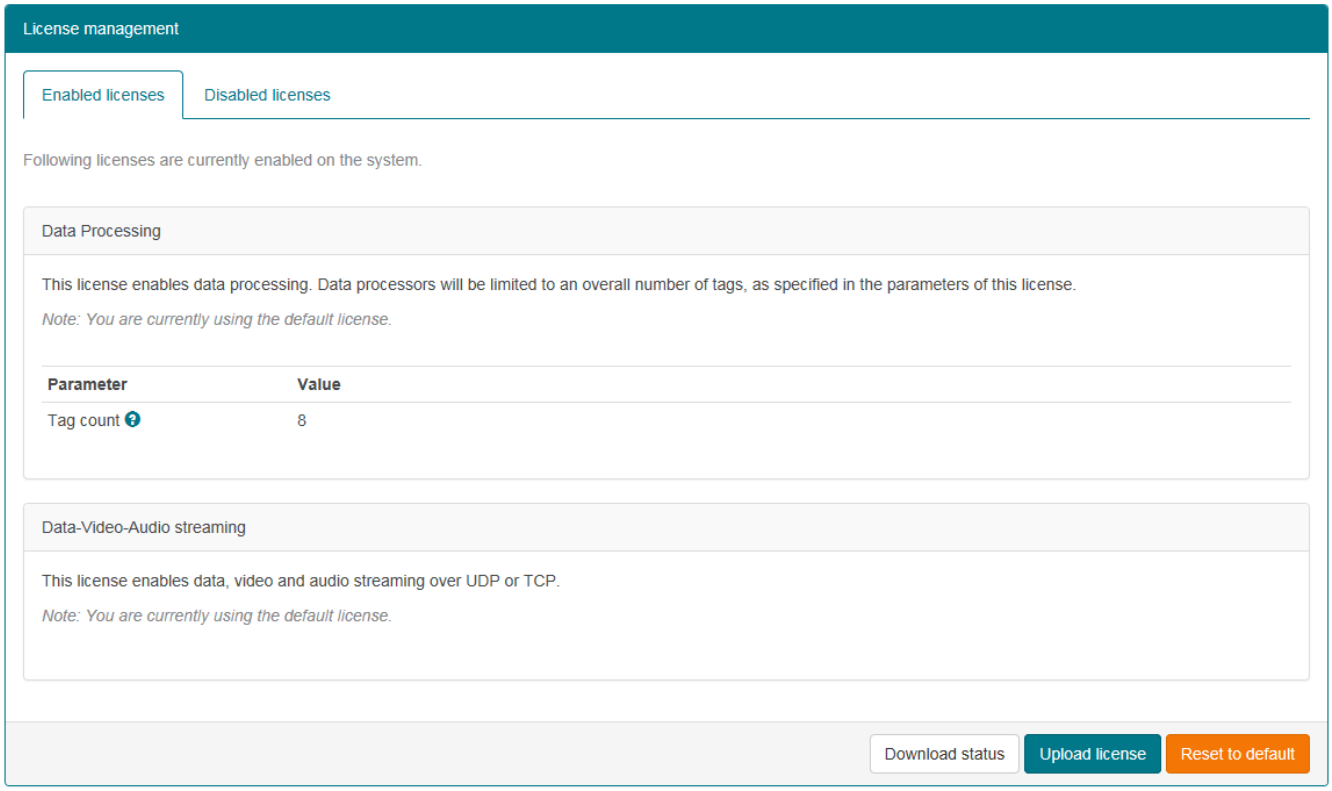


Fill in the form and click on "Update".



17 License system

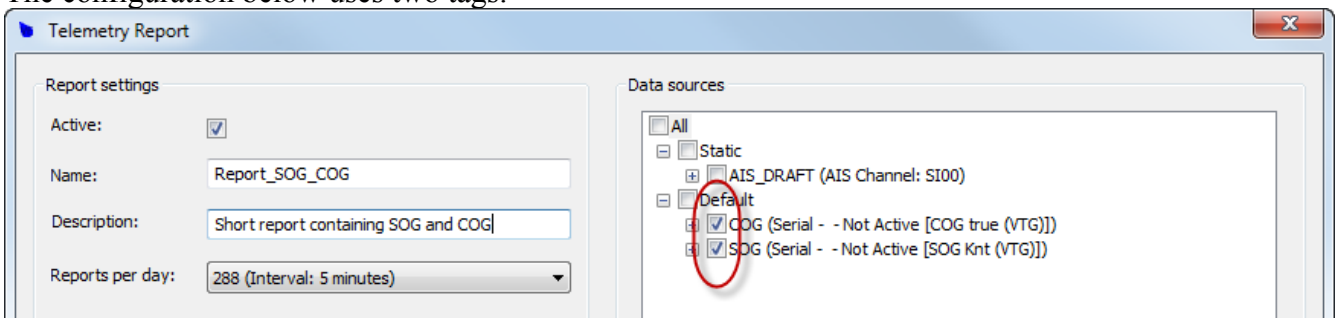
Some functionality in the VRS may require a license.
Go to “System->License”.



17.1 License for data processors

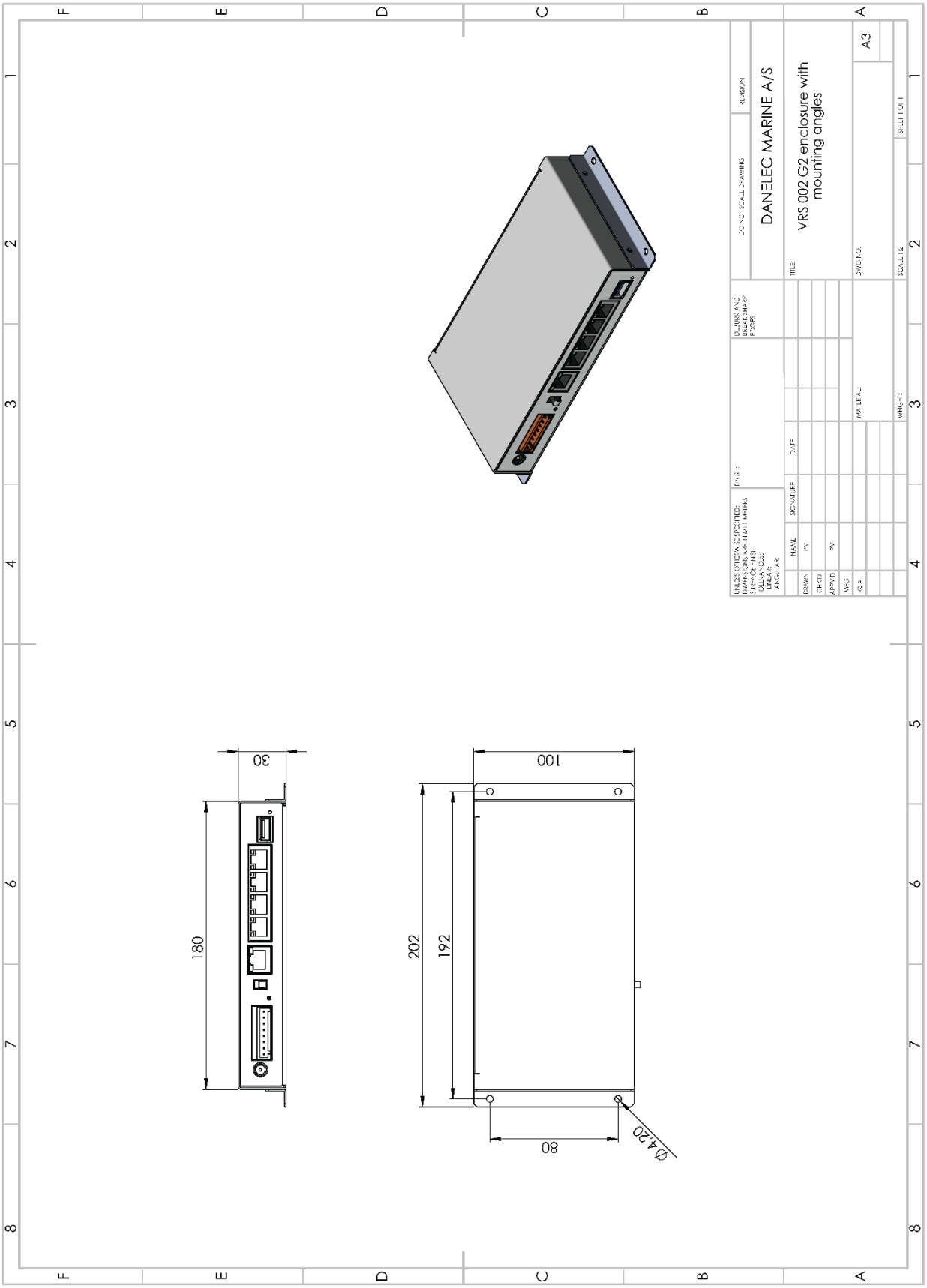
The number of tags used by the data processors (combined) is, per default, limited to eight.

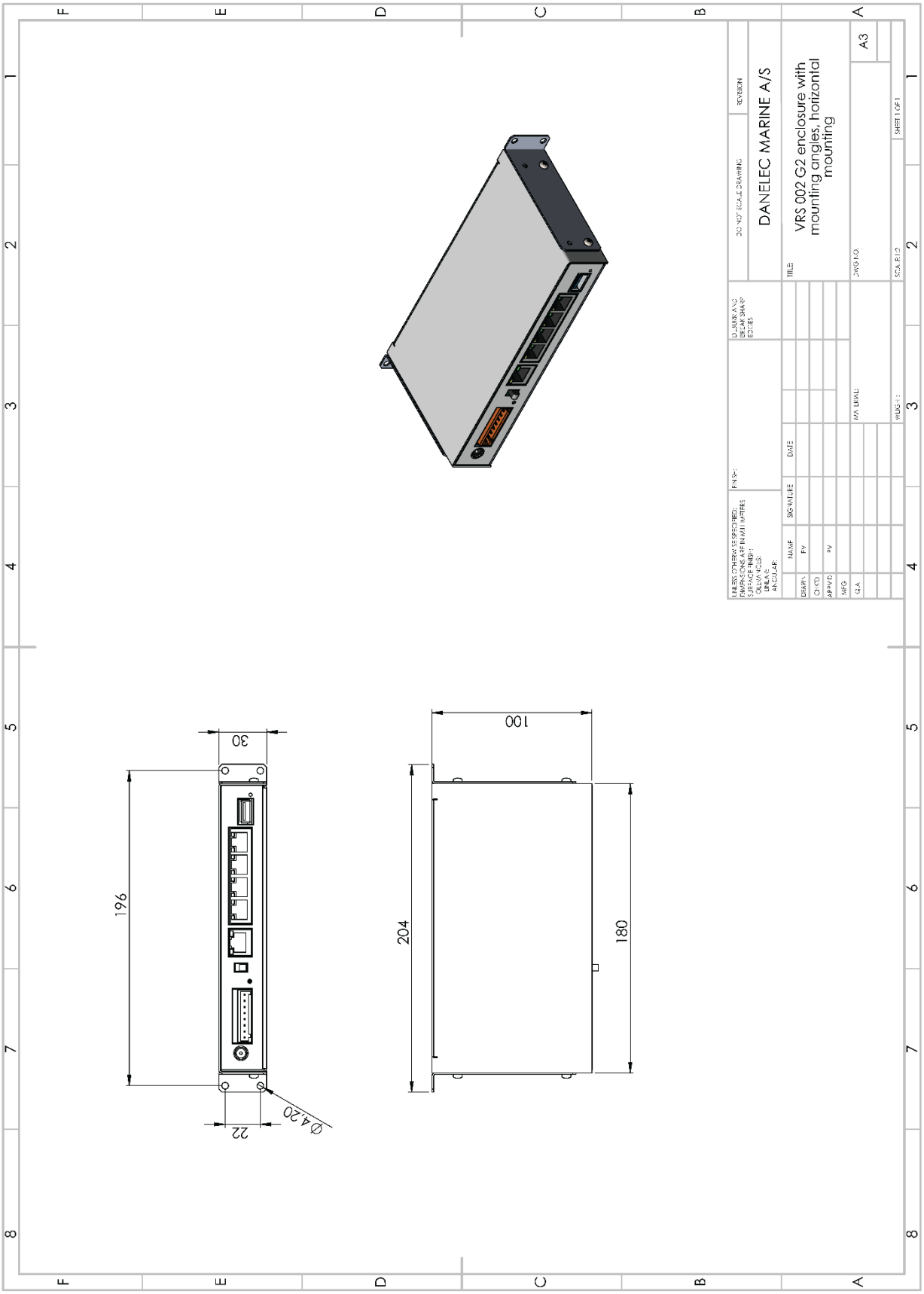
Tags are “allocated” when a data processor configuration is created; see section 5.2.3.
The configuration below uses two tags.



It is possible to purchase a license that will change the limitation on available tags. A license is distributed as a file and must either be uploaded manually (System->License) or from an administrative server, see section 8.4.

18 Installation drawings




DO NOT SCALE DRAWING
REVISION
DANELEC MARINE A/S
VRS 002 G2 enclosure with mounting angles, horizontal mounting
TITLE:
A3
DWG NO.
SCALE: 1:2
SHEET 1 OF 1
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETRES SURFACE FINISH: DEBURR ALL EDGES BREAK ALL SHARP EDGES ANGULAR:
FINISH:
DEBURR AND BREAK SHARP EDGES
NAME
SIGNATURE
DATE
DRAWN
CHKD
APPVD
MFG
Q.A.
MATERIAL
WEIGHT:
30
196
22
Ø4.20
100
204
180

19 Appendix A – Digital signatures

The VRS can optionally be configured to use digital signatures for signing reports and/or configurations subject to remote update. This appendix describes how to verify and create these digital signatures.

19.1 Verification of report signature

When the VRS is configured to use digital signatures on reports for a destination, then each report uploaded to the destination will be accompanied by a separate file containing a digital signature for the report. It is possible to authenticate the received report on the destination server by verifying the signature using the public key downloaded from the VRS as described in section 12.2.

The signature can be verified by any 3rd party application that can verify SHA1 signatures. Below is an example of verification using the openssl (ver. 1.0 or higher) open source application:

```
openssl dgst -sha1 -verify vrs-reports.public.pem -signature report_file.sign
report_file
```

where:

vrs-reports.public.pem is the public key downloaded from the VRS.
report_file.sign is the signature file.
report_file is the actual report to be authenticated

Running the above command will produce a verdict: “Verified OK” or “Verification Failure”.

Below is an example of a script that can automate the report authentication process:

```
#!/bin/sh

# INFO: Sample script for reports signature verification

##############################################################
# USER CONFIGURATION
##############################################################

# (paths can be absolute or relative to the script file)

# directory path where new reports are uploaded
NEW_REPORTS_DIR="new_reports"

# directory path where verified reports are moved
VERIFIED_REPORTS_DIR="verified_reports"

# directory path where reports that failed to be verified are moved
INVALID_REPORTS_DIR="invalid_reports"

# command name or full path to openssl
# (if absolute path - other directories paths also shlould be)
OPENSSL_CMD="openssl"

# path to public key used for signature verification
PUBLIC_KEY="vrs-reports.public.pem"
```

```
# file where errors will be logged
LOG_FILE="log.txt"
############################################################

# checking for new reports

# make sure that directories are created
mkdir -p $NEW_REPORTS_DIR
mkdir -p $VERIFIED_REPORTS_DIR
mkdir -p $INVALID_REPORTS_DIR

# for every xml, gz or bz2 file in NEW_REPORTS_DIR directory
find $NEW_REPORTS_DIR -name "*.xml" -o -name "*.gz" -o -name "*.bz2" | while
read file; do
    echo "Processing file '$file'"

    # signature file name should end with ".sign"
    signature_file=$file".sign"

    # verify digital signature of the report
    $OPENSSL_CMD dgst -sha1 -verify $PUBLIC_KEY -signature $signature_file $file

    if [ $? -eq 0 ] ; then
        # verification succeed - move report to 'verified' directory
        mv -f $file $VERIFIED_REPORTS_DIR
        mv -f $signature_file $VERIFIED_REPORTS_DIR
    else
        # verification failed - move report to 'invalid' directory
        mv -f $file $INVALID_REPORTS_DIR
        mv -f $signature_file $INVALID_REPORTS_DIR
        # log event this event in file
        echo "Failed to verify file: $file" >> $LOG_FILE
    fi
done
```

19.2 Signing a configuration for remote update

If the VRS is configured to use digital signature on configurations for remote, update then the configuration file must be signed and the .signature file must be located in the same directory as the configuration file.

Signing the configuration file requires a private key and a corresponding public key. Please see sample script below for information on how to create a PKI key pair. Use the private key to create a signature file e.g. using openssl:

```
openssl dgst -sha1 -sign vrs-configs.private.pem -out config_file.signature
config_file
```

where :

vrs-configs.private.pem is the private key.
config_file.signature is the signature file.
config_file is the actual configuration to be signed.

Running the above command will produce a file containing the signature.

Below is an example of a script that can automate the configuration signing process:

```
#!/bin/sh

# INFO: Sample script demonstrating how to sign a configuration
# Usage: sign_configuration.bat <config_file>
#                       output signature file in <config_file>.signature

##############################################################
# USER CONFIGURATION
##############################################################

# command name or full path to openssl
OPENSSL_CMD="openssl"

# full path to private key used for signing
PRIVATE_KEY="private_key.pem"

# full path to public key used for verification
PUBLIC_KEY="public_key.pem"
##############################################################


# Below is an example how to generate key pair used for signing (openssl ver >=
1.0 ):
# ----
# openssl genpkey -algorithm RSA -out private_key.pem -pkeyopt
rsa_keygen_bits:2048
# ----
# This generate a private key file (also contains public key) that is used for
signing.
# Make sure that is stays secure - do not share it with anyone untrusted!
# Now we have to extract the public key used in signature verification:
# ----
```

```
# openssl rsa -pubout -in private_key.pem -out public_key.pem
# ----
# The file public_key.pem should be uploaded to the corresponding “Destination”
on VRS


if [ -z "$1" ]
  then
    echo “Please provide path to config file as first argument...”
    exit 1
fi

config_file=$1
signature_file="$config_file.signature"

# sign configuration using private key - output in <config_file>.signature
$OPENSSL_CMD dgst -sha1 -sign $PRIVATE_KEY -out $signature_file $config_file

# verify signature
$OPENSSL_CMD dgst -sha1 -verify $PUBLIC_KEY -signature $signature_file
$config_file
```